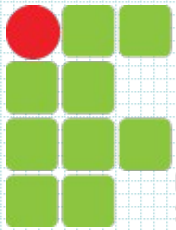
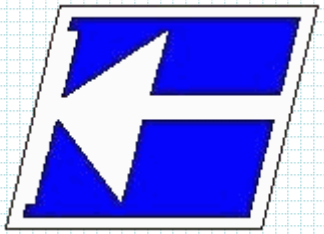
Instituto Federal de Educação, Ciência e Tecnologia de Santa Catarina
Departamento Acadêmico de Eletrônica
Curso de Graduação em Engenharia Eletrônica





Fontes de Alimentação Chaveadas

Prof. Joabel Moia.

Florianópolis, setembro de 2026.

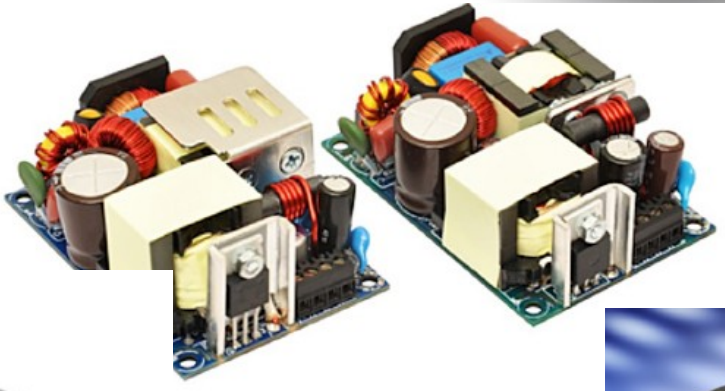
Fontes Chaveadas

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Potência:











Fontes Chaveadas

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Necessidade de Fontes de Tensão Contínua:

- Obtenção de tensões contínuas a partir de tensões alternadas, que é o tipo de tensão disponibilizada pelas concessionárias de energia.
- Normalmente são empregadas em circuitos eletrônicos.
- São utilizadas as mais diversos tipos de tecnologia para o processamento de energia de corrente alternada para corrente contínua.
- As tecnologias vão desde as mais simples (fontes lineares), até as mais complexas (chaveadas).

Fontes Chaveadas

A importância da eletrônica - Aplicação:

1. Telecomunicações e entretenimento;
2. Computadores e calculadoras;
3. Sistemas de controle automático;
4. Instrumentação;
5. Eletrônica automotiva;
6. Geração e distribuição de energia;
7. Radar;
8. Circuitos integrados;
9. Entre outros ....





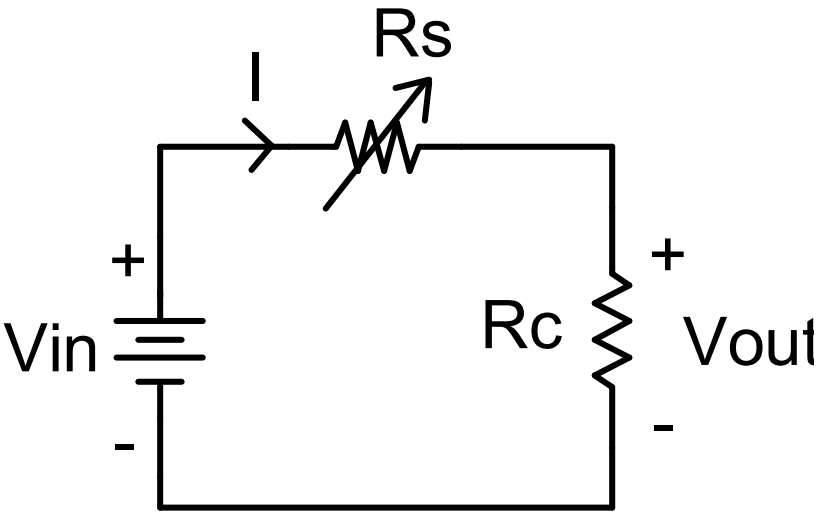




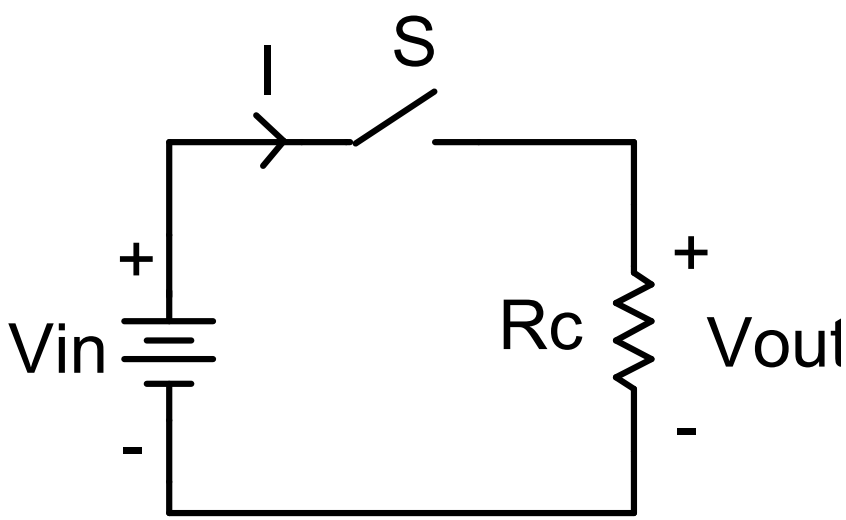
Fontes Chaveadas

Fontes de tensão lineares e chaveadas:

- As fontes lineares convertem a tensão alternada da rede em tensões contínuas, normalmente de baixa amplitude, sem o uso de componentes chaveados (comutados);
- Fontes chaveadas exercem a mesma função, mas utilizando componentes comutados (chaveados).



Regulador linear



Regulador chaveado

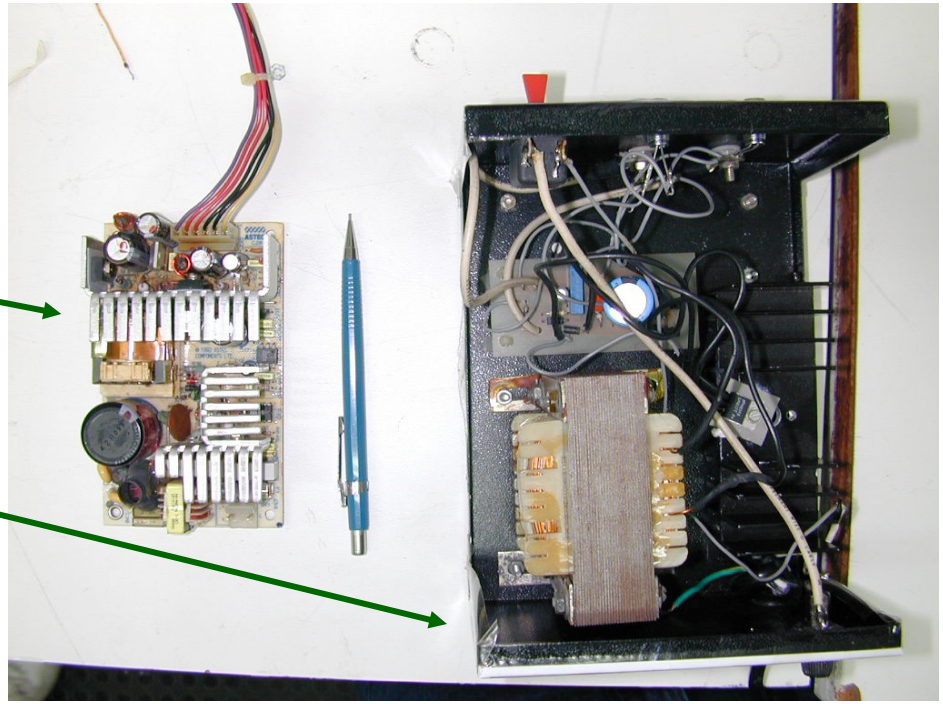
Fontes Chaveadas

Fontes de tensão lineares x chaveadas:

- Fontes lineares: são mais robustas, simples e fáceis de projetar, geralmente são mais baratas, são muito volumosas e pesadas. Menor eficiência
- Fontes chaveadas: não são tão robustas, mais difíceis de projetar e **consertar**, geralmente são mais caras, são pequenas e leves. Maior eficiência.

Fonte chaveada de 65 W

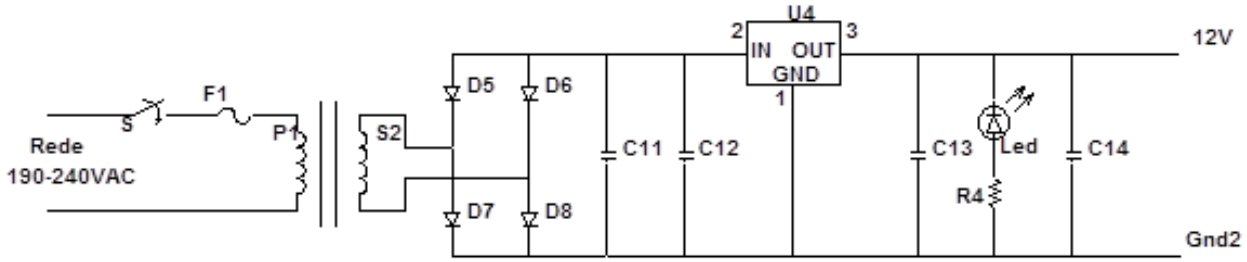
Fonte linear de 29 W



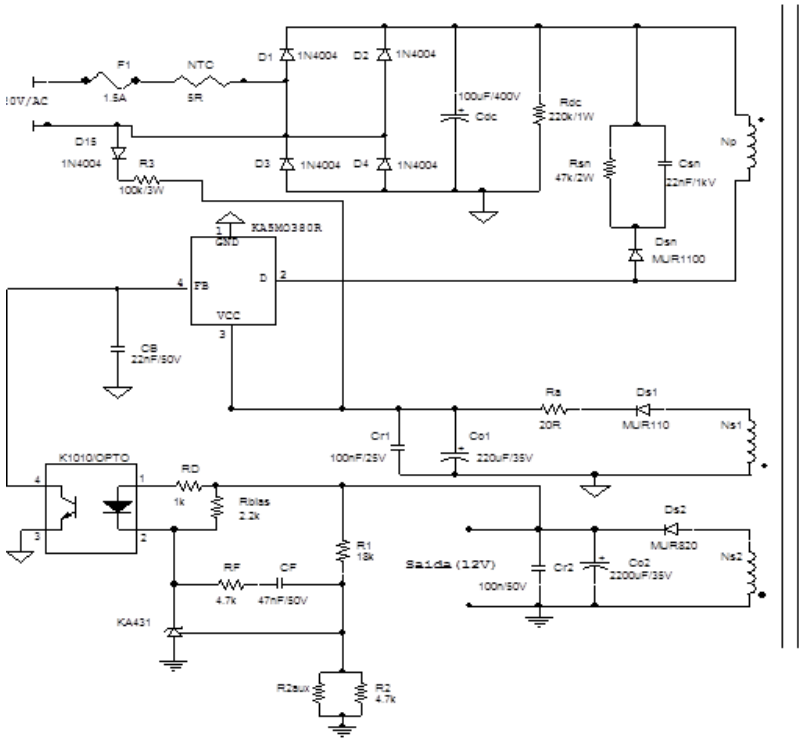
http://www.priberam.pt/

Fontes Chaveadas

Fonte Linear – Menos Complexa:



Fonte Chaveada – Mais Complexa:

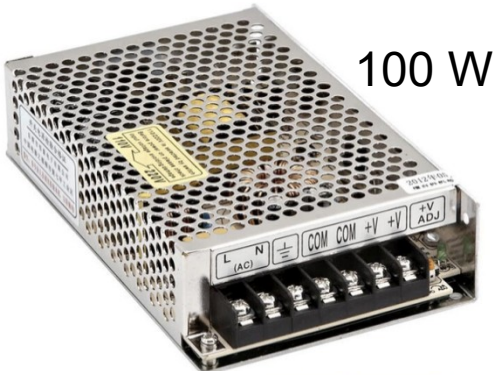


Fontes Chaveadas

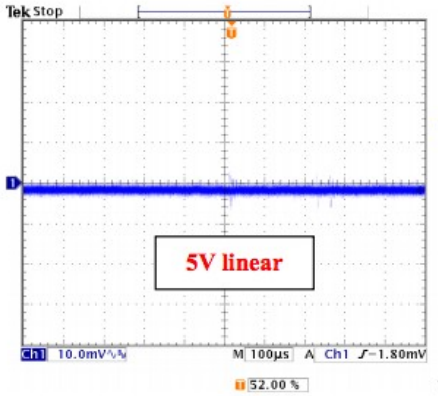
Fonte linear versus comutada

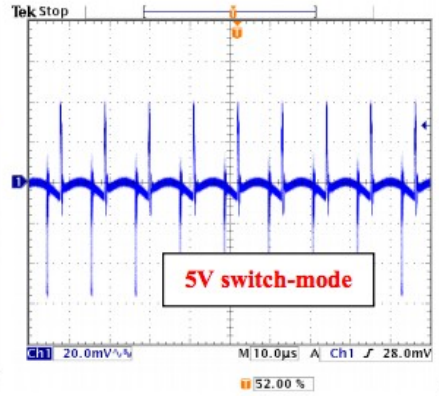


90 W



100 W





	Linear	Switch-mode	Comments
Size	✖	✓	Typically 80% smaller
Weight	✖	✓	Typically 80% lighter
Input Voltage Range	✖	✓	10% vs. up to 300% range
Efficiency	✖	✓	Calculate it long term!
Reliability	✓	✖/✓	Component count method, demonstrated probably equal
Ripple & Noise	✓	✖/✓	Up to 10,000 times - often possible to overcome though
Transient Response	✓	✖	Up to 100 times - necessary in specialized areas
Low leakage Current	✓	✖/✓	Often used in medical systems, switch-mode gaining share

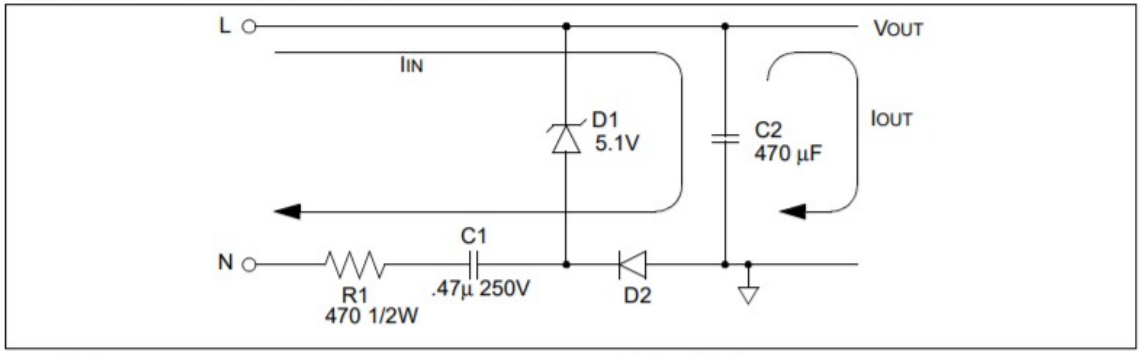
https://www.valuetronics.com/Manuals/Lambda_%20linear_versus_switching.pdf

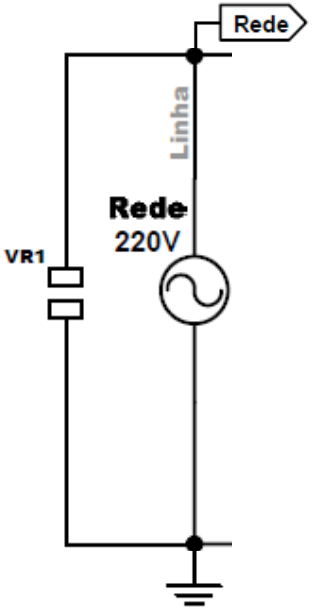
Fontes Chaveadas

Fontes de tensão lineares baixa potência sem transformador:

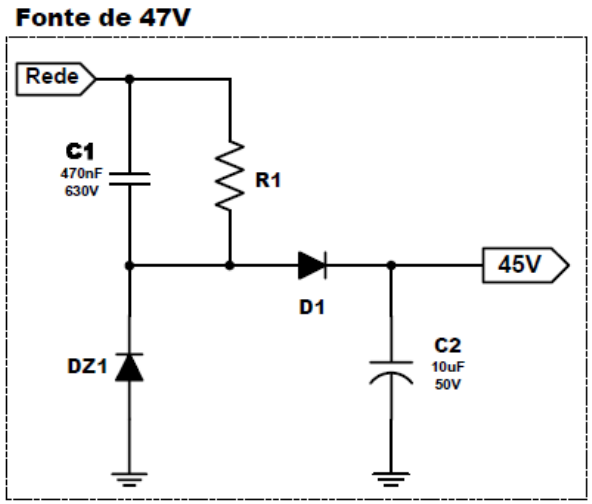
AN954 - Transformerless Power Supplies: Resistive and Capacitive (Microchip)

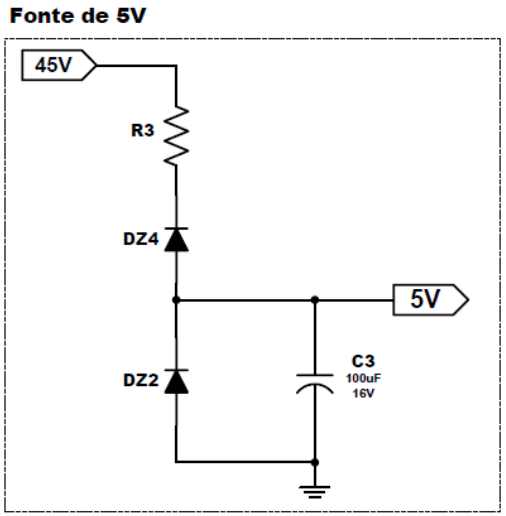
FIGURE 1: CAPACITIVE POWER SUPPLY





Exemplo Produto:

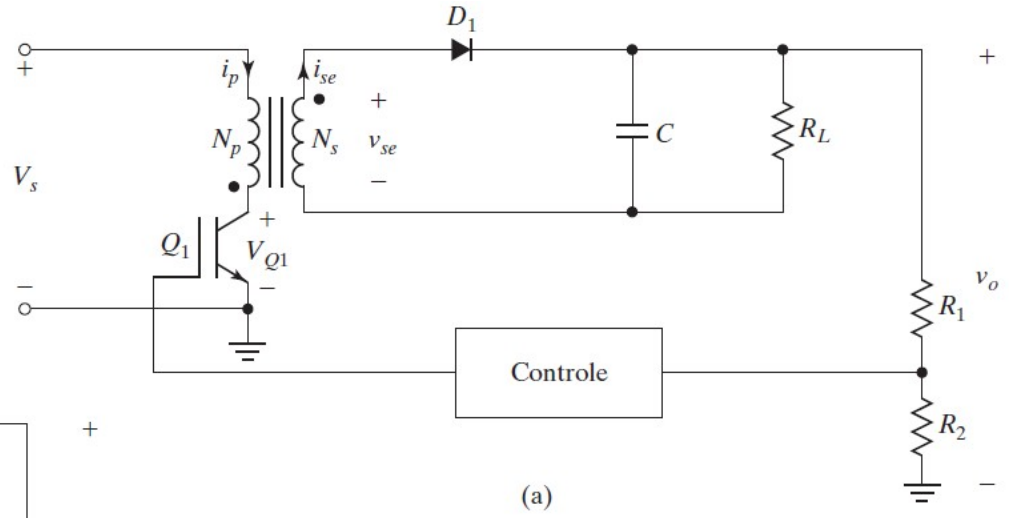


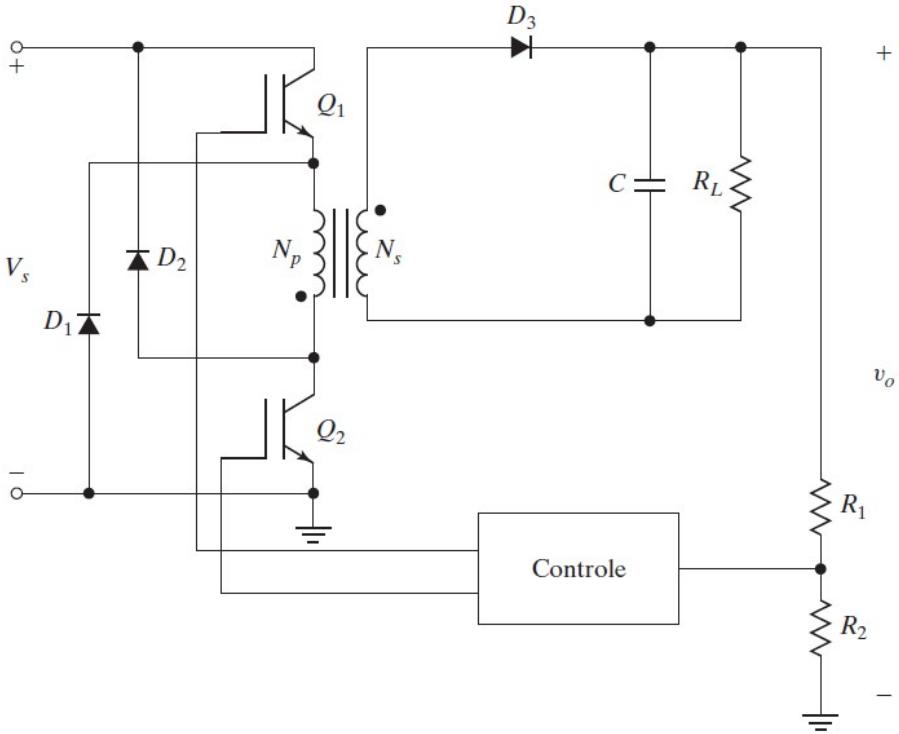


Fontes Chaveadas

Principais Topologias Empregas em Fontes Chaveadas Isoladas:

Conversor Flyback



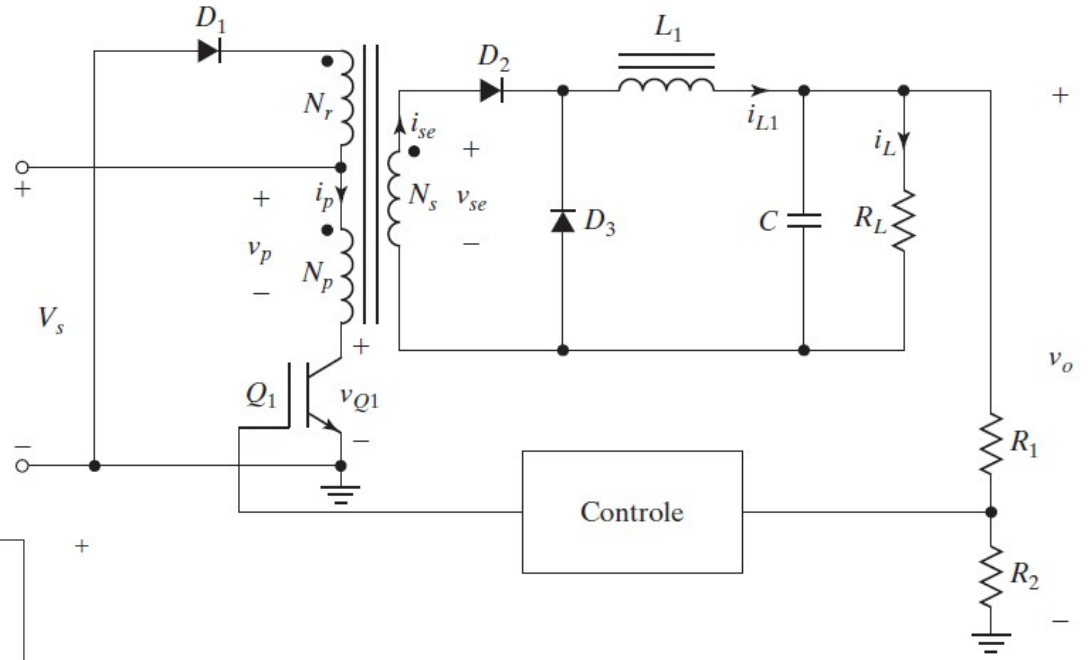


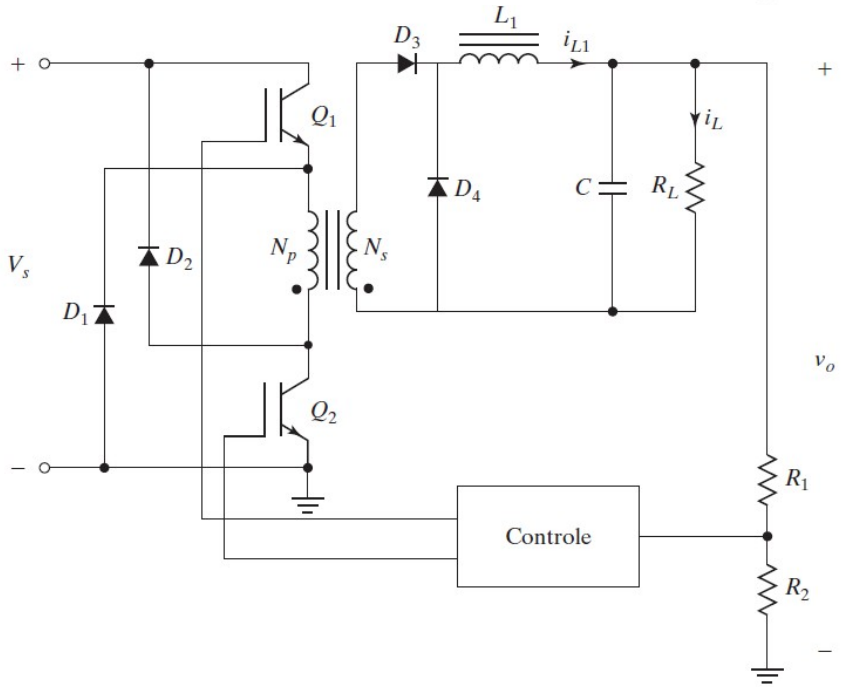
Conversor Flyback duas Chaves

Fontes Chaveadas

Principais Topologias Empregas em Fontes Chaveadas Isoladas:

Conversor Forward



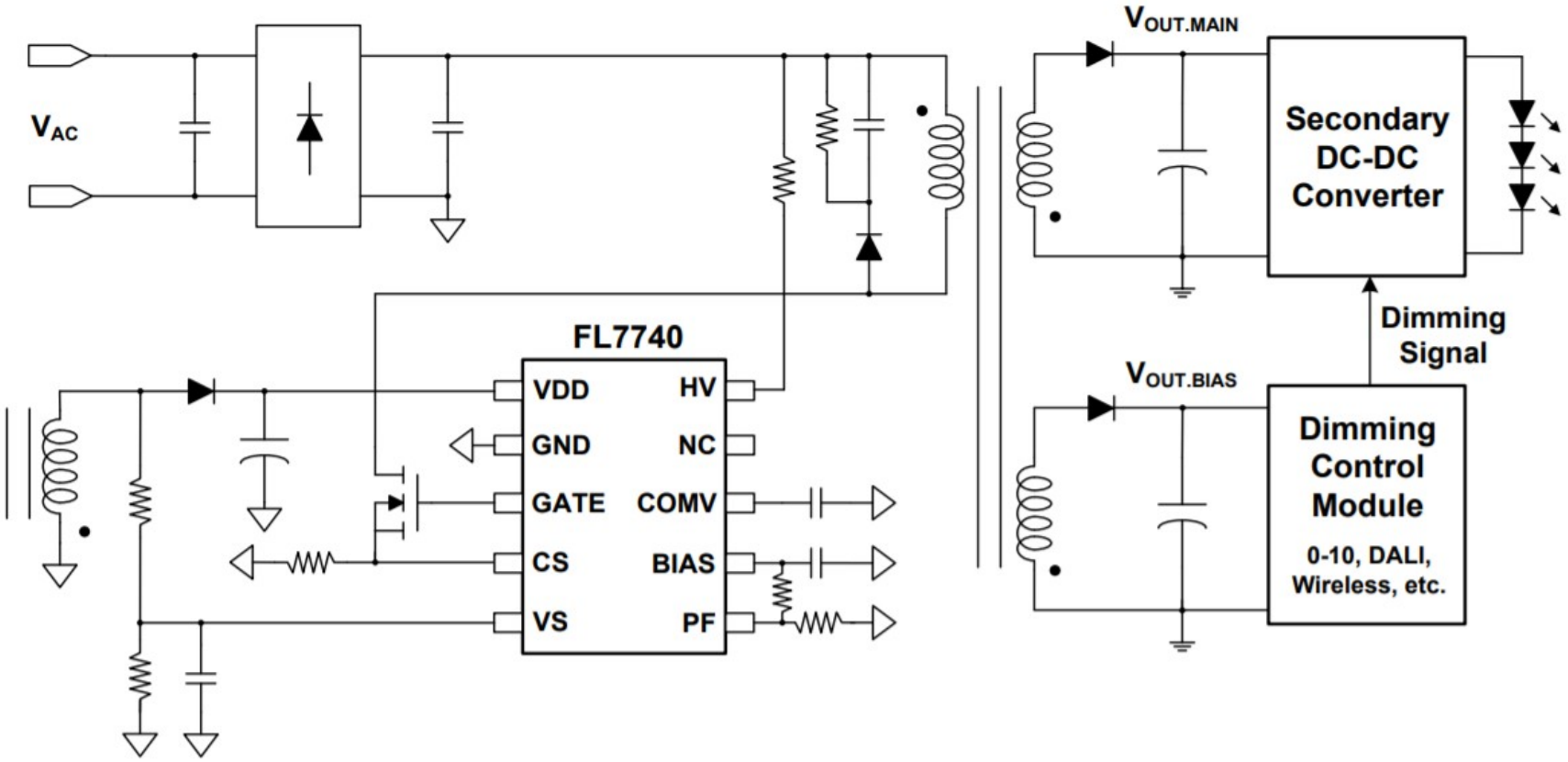


Conversor Forward duas Chaves

Fontes Chaveadas

Estágio de entrada de uma fonte chaveada:

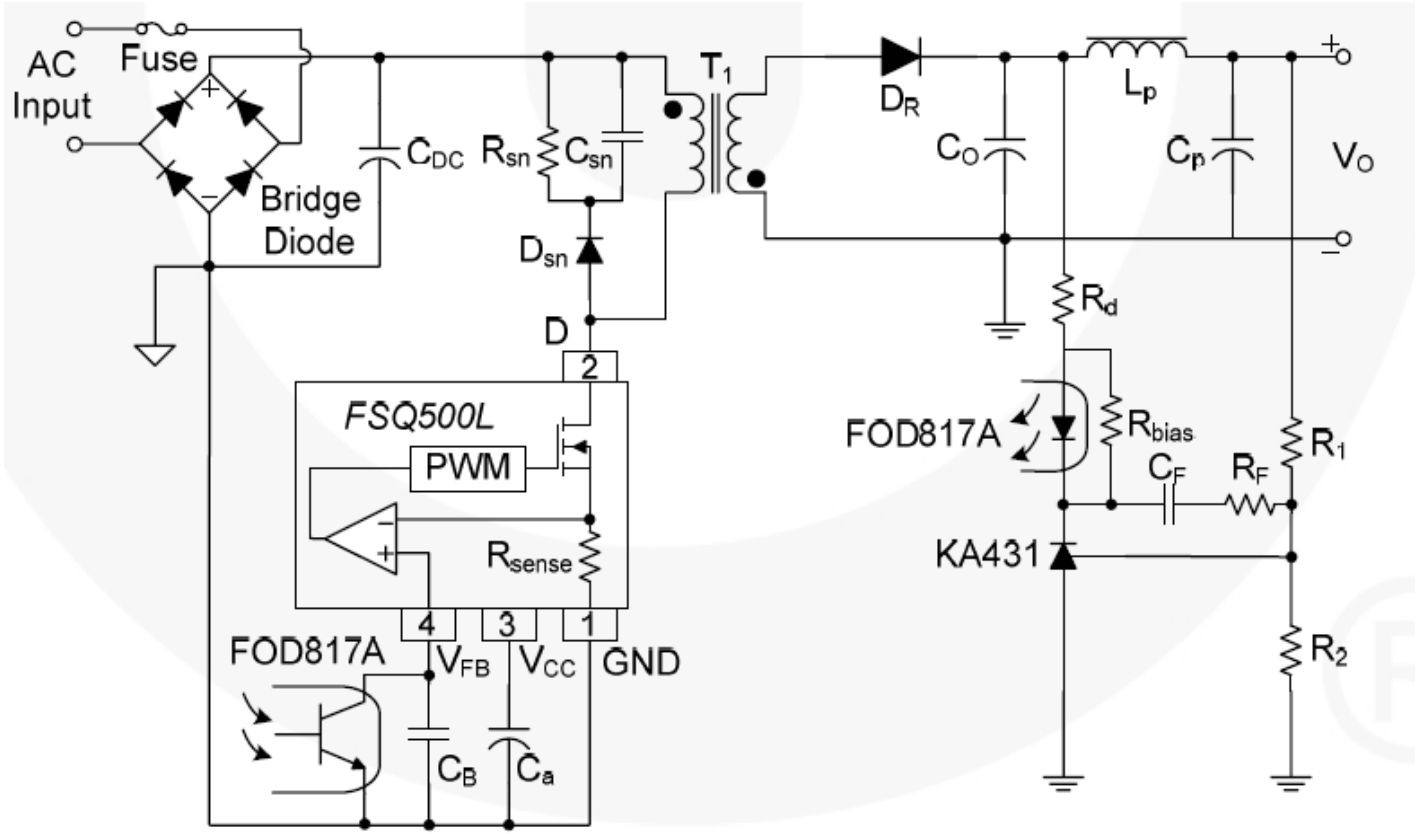
1. Retificador;
2. Filtro capacitivo;
3. Conversor CC-CC Isolados (Conversor Forward ou Flyback).



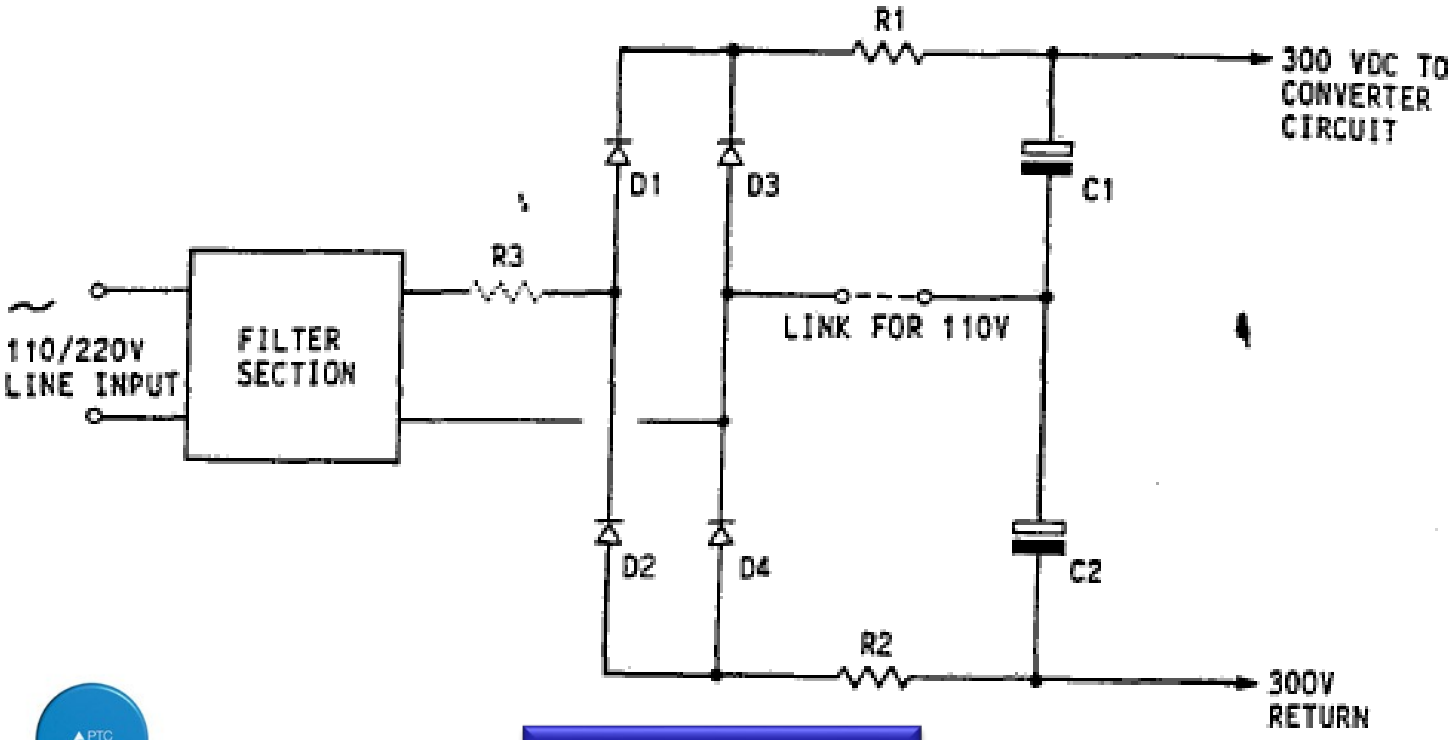
Fontes Chaveadas

Estágio de entrada de uma fonte chaveada:

1. Retificador;
2. Filtro capacitivo;
3. Conversor CC-CC Isolados (Conversor Forward ou Flyback).



Partida do estágio de potência



Resistores série



Varistor



Fontes Chaveadas – Fusíveis

Fusíveis de ação lenta:

- Vidro;
- Areia;
- Cerâmica.

Fusíveis de ação rápida:

- Vidro;
- Areia;
- Cerâmica.

Resistores fusíveis.

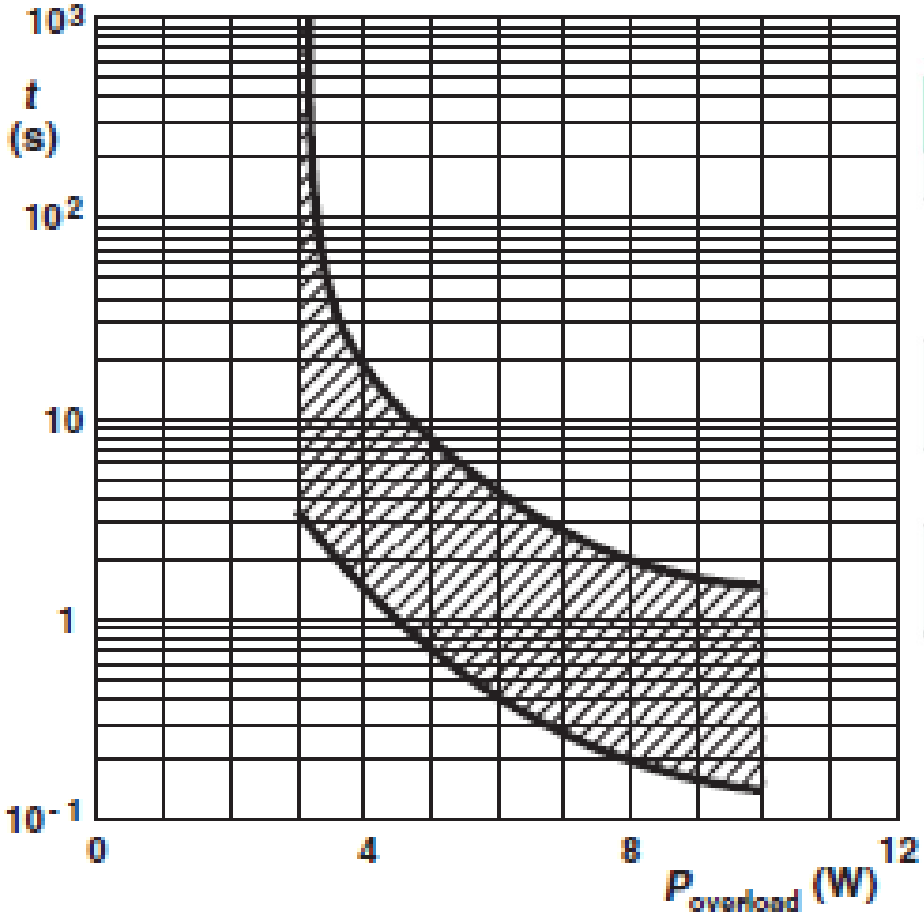








Fontes Chaveadas – Fusíveis



http://www.vishay.com

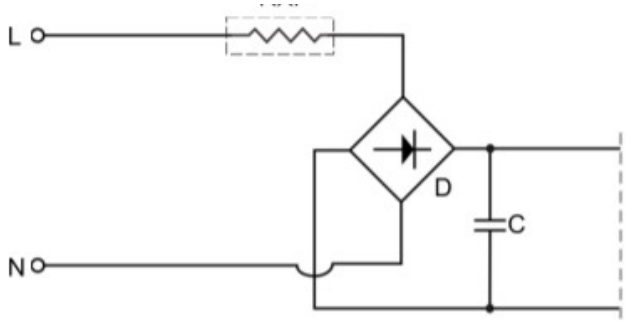
% of Ampere Rating	Ampere Rating	Opening Time
100%	0.062A - 15A	4 Hours, Min.
200%	0.062A - 7A	1 Second, Max.
	10A	3 Seconds, Max.
	12 - 15A	10 Seconds, Max.
275%	0.500A, 1A, 2A, 2.5A, 3A, 4A, 5A, 7A, 10A	300 msecs., Max.
400%	0.5A, 1A, 2A, 2.5A, 3A, 4A, 5A, 7A, 10A	30 msecs., Max.
1000%	0.500A, 1A, 2A, 2.5A, 3A, 4A, 5A, 7A, 10A	4 msecs., Max.

http://www.littelfuse.com

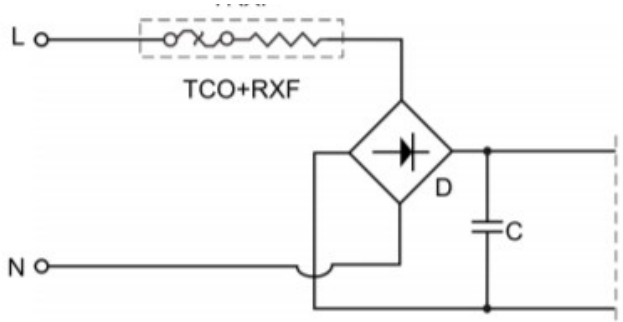
Fontes Chaveadas – Fusistor

Fusistor, Resistor Fusível ou Resistor Fusível Térmico:

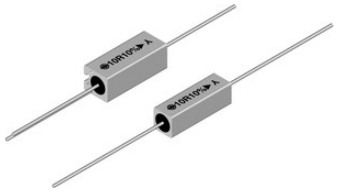
- Um fusistor é um resistor de baixo valor colocado de tal maneira que ele funciona como um fusível. Quando a corrente aumenta no circuito, a resistência aumenta, diminuindo a corrente.



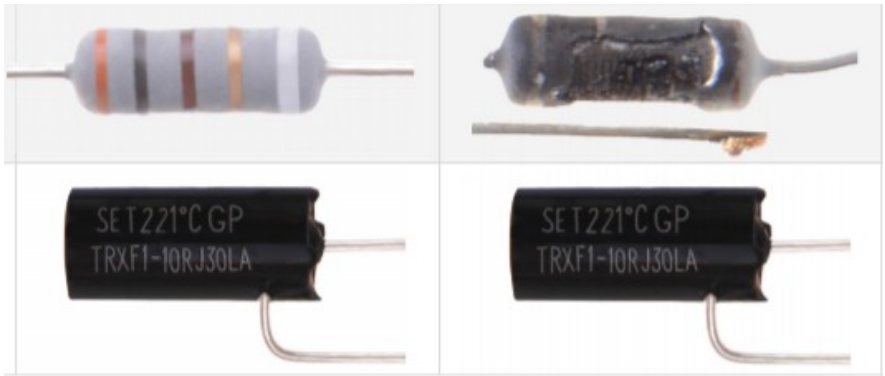
Resistor Fusível



Resistor Fusível Térmico









Fusível

- Corrente de ruptura: É a responsável por estipular o valor máximo de corrente que o fusível consegue suportar antes que se funda;

- Corrente nominal: Essa corrente mostra o quanto o componente é capaz de suportar sem haja rompimento.

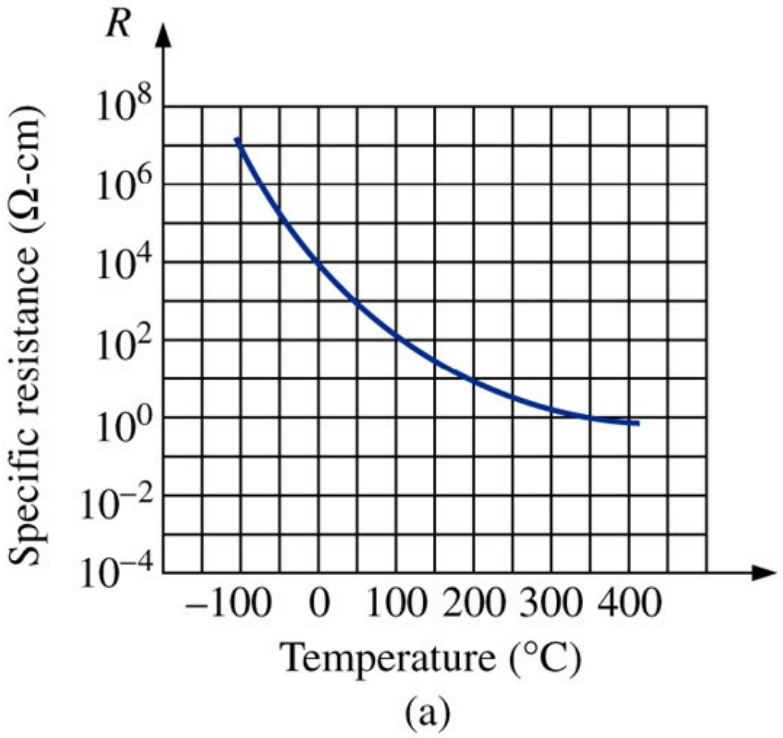
Como especificar:

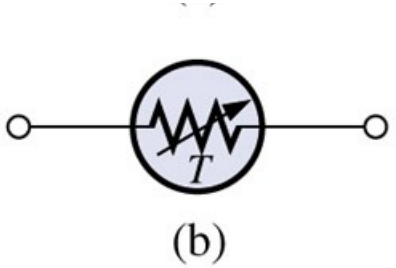
1) Calcular a corrente nominal na condição de maior potência do circuito eletrônico. Levar em consideração a variação de tensão de alimentação do circuito eletrônico;
2) Levar em consideração se o circuito equipamento tem corrente de partida (energização) para escolher a velocidade de atuação do fusível;
3) Multiplicar o valor da corrente nominal calculada por 2x a 3x (outros valores podem ser analisado, dependendo do tempo de resposta do fusível, a corrente de ruptura e nominal).

Fontes Chaveadas – Termistores

Termistor:

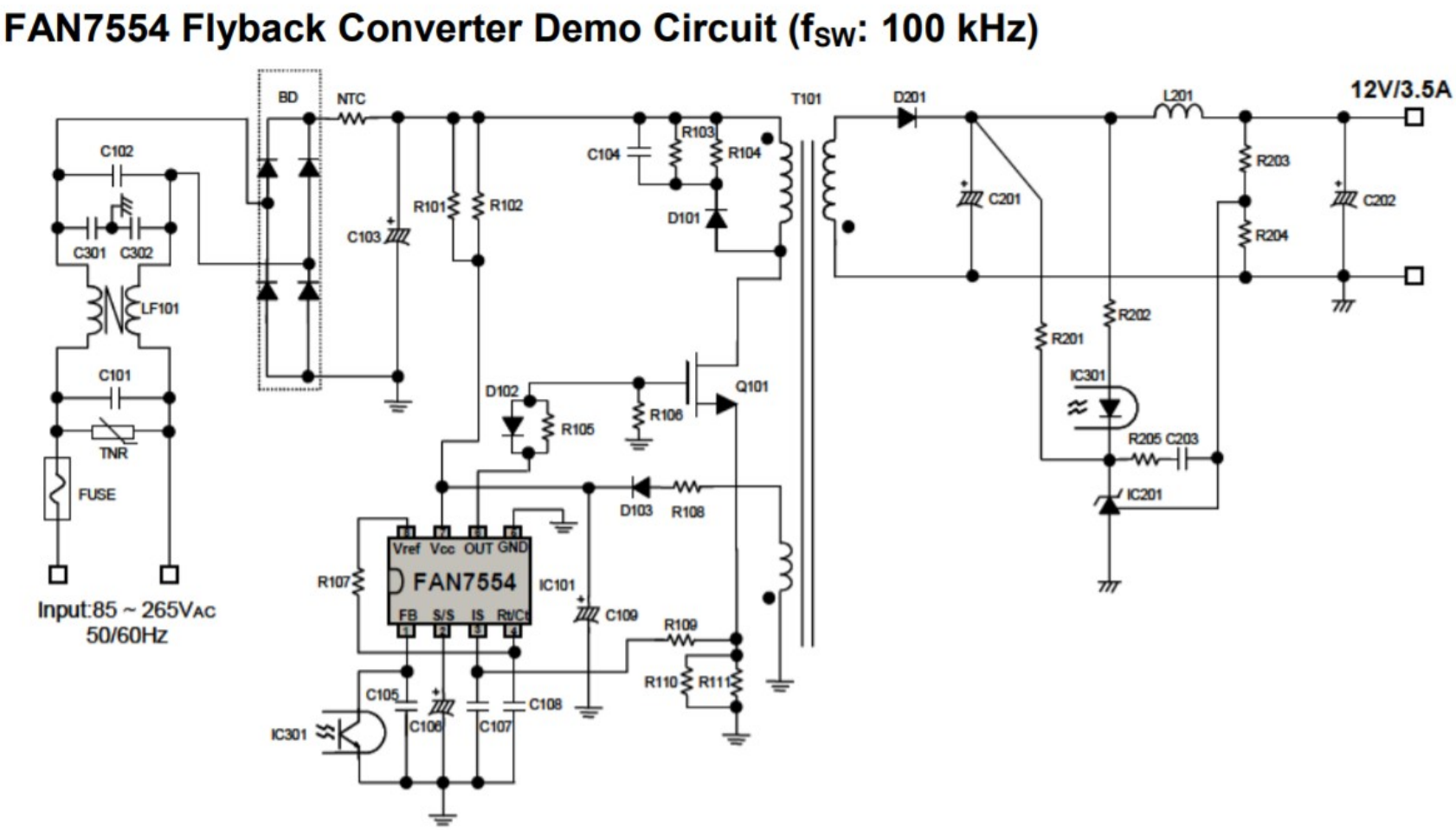
- Resistor cuja resistência é sensível à variação da temperatura.





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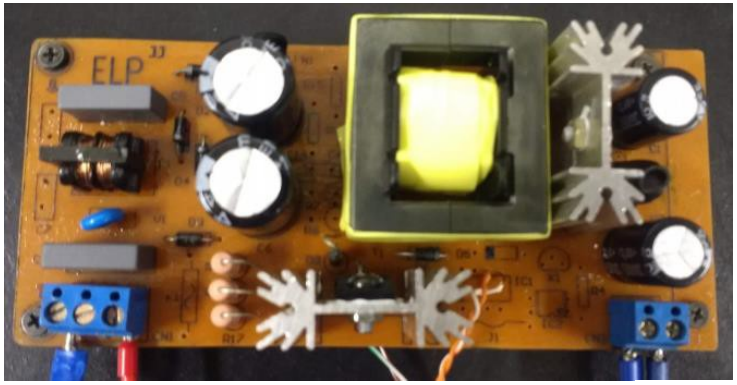
Estágio de conversão CC-CC de uma fonte chaveada:



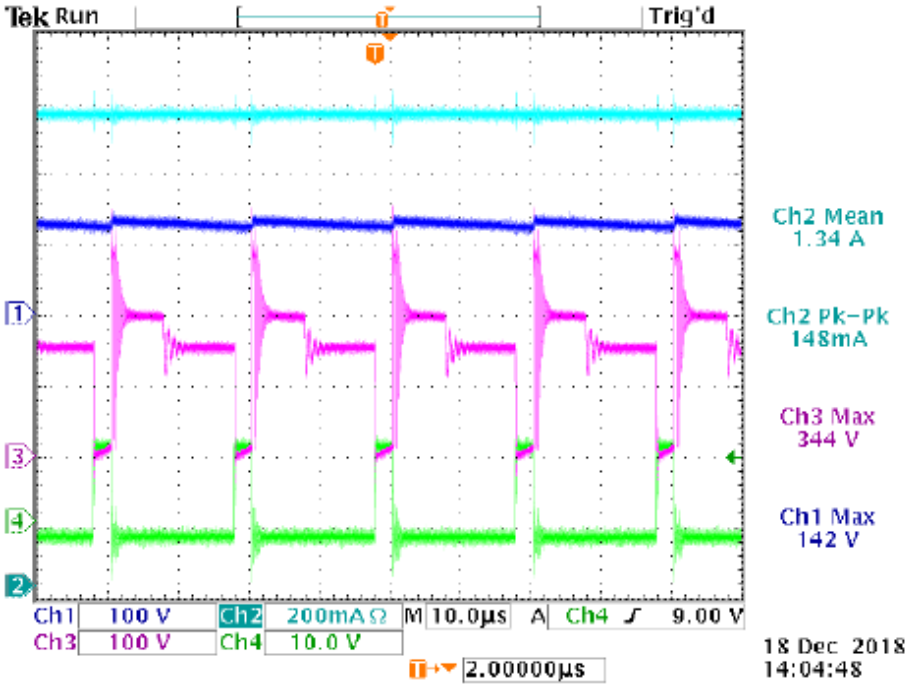
Fontes Chaveadas

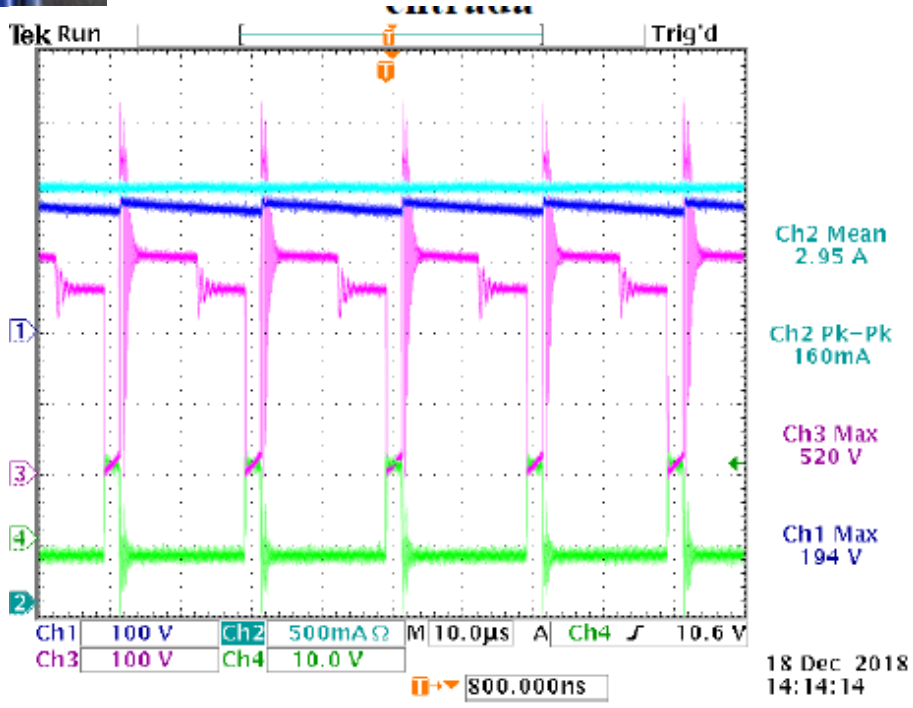
Exemplo de uma fonte chaveada:

Semestre 2018-2: João Cardoso e Jonas Vieira Souza



FAN7554 – Flyback:
Tensão de Saída: 15V
Corrente Máxima: 5 A





Fontes Chaveadas

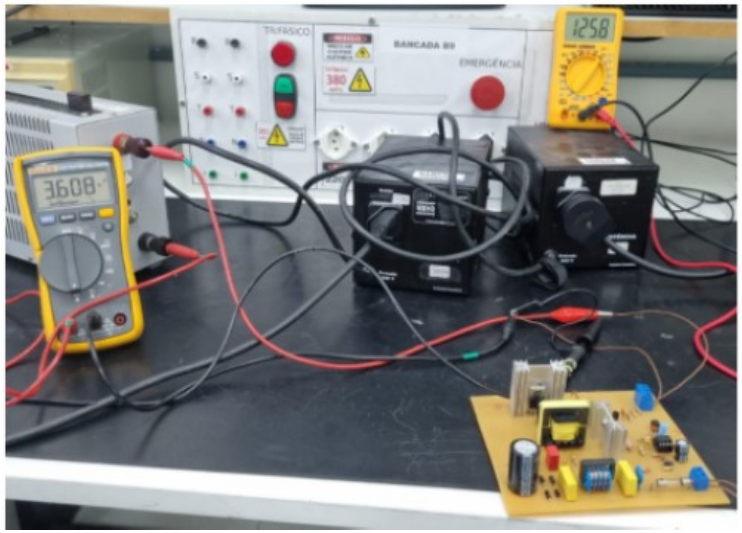
Exemplo de uma fonte chaveada:

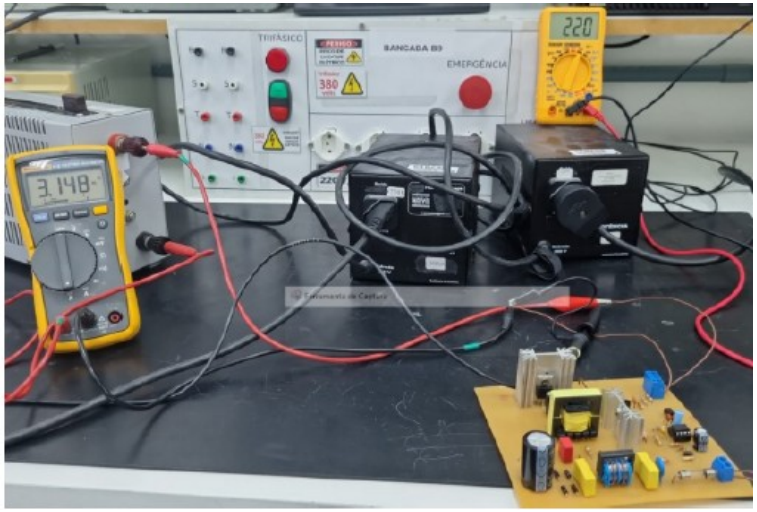
Semestre 2024-2: Alexsander Vieira, Bruno C. Mazzochi e Ueslei Marian

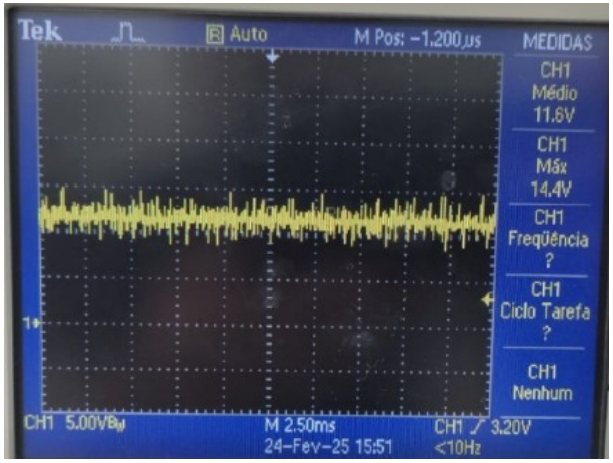
FAN7554 – Flyback:

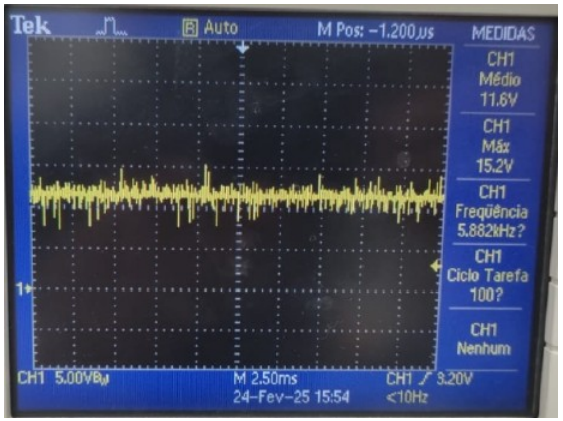
Tensão de Saída: 12 V

Corrente Máxima: 3,5 A



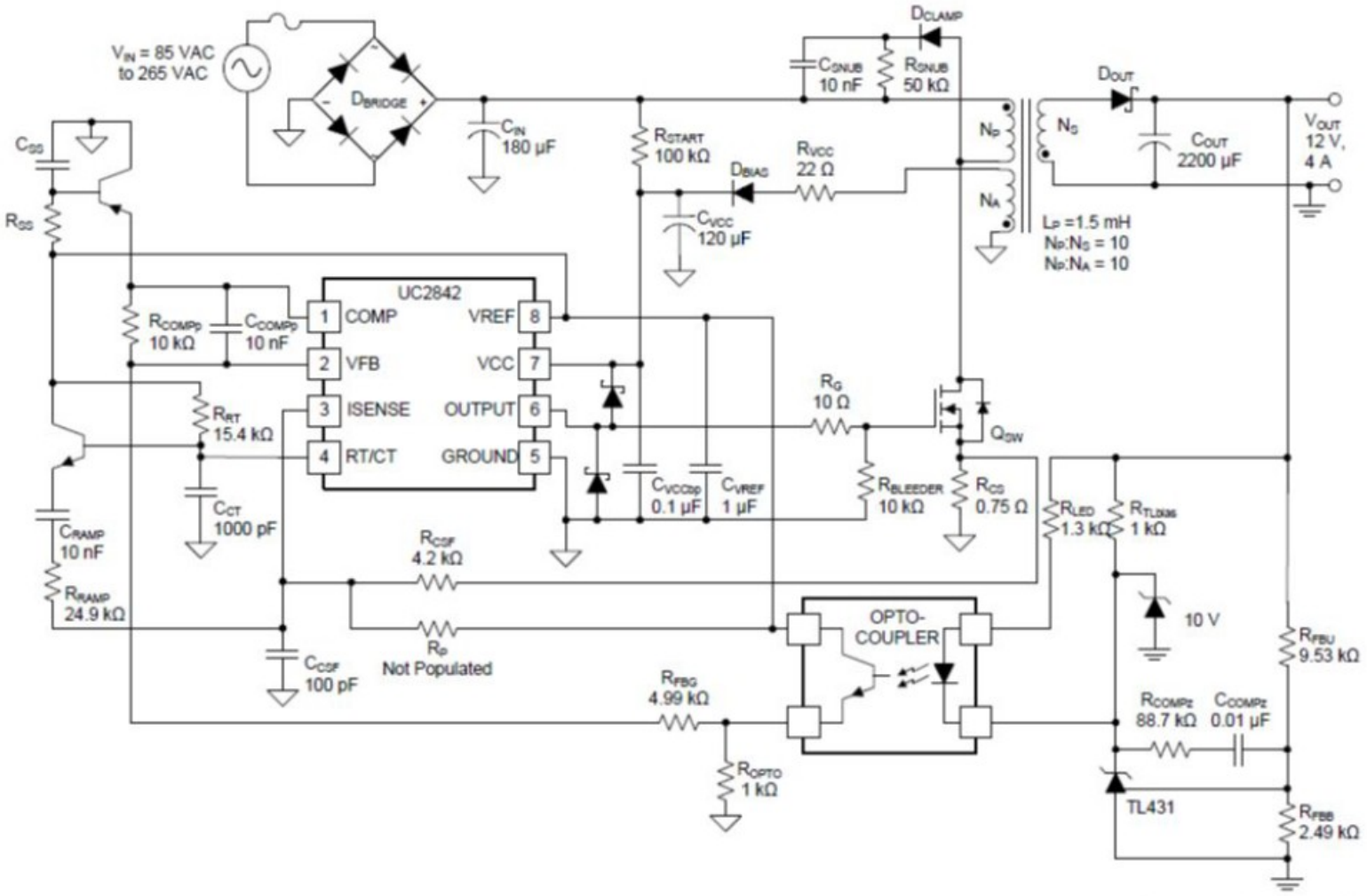






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Estágio de conversão CC-CC de uma fonte chaveada: UCx84x

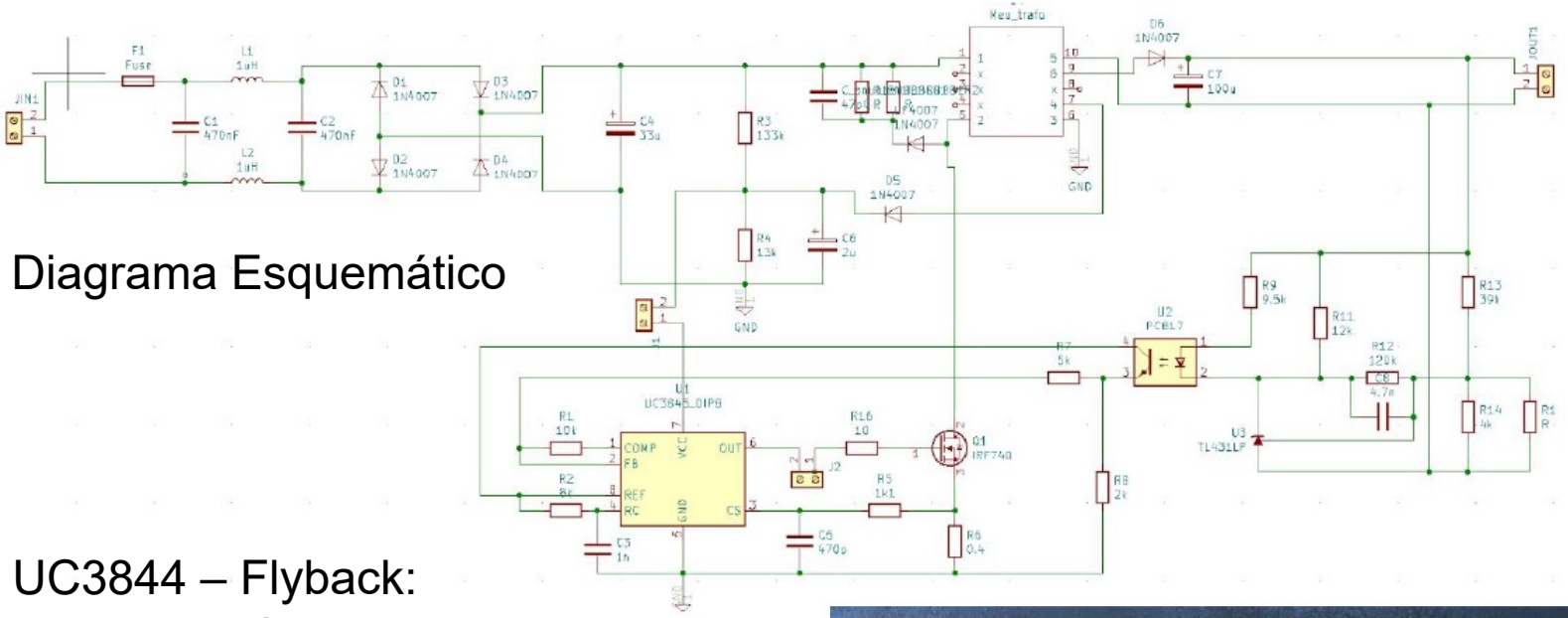


Fontes Chaveadas

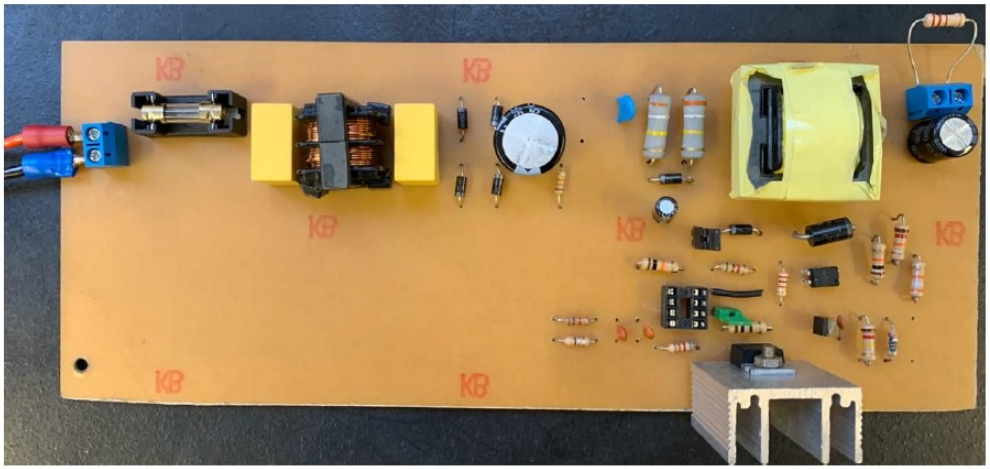
Exemplo de uma fonte chaveada:

Semestre 2019-2: Bruna Martini e Rafael Reis

Diagrama Esquemático



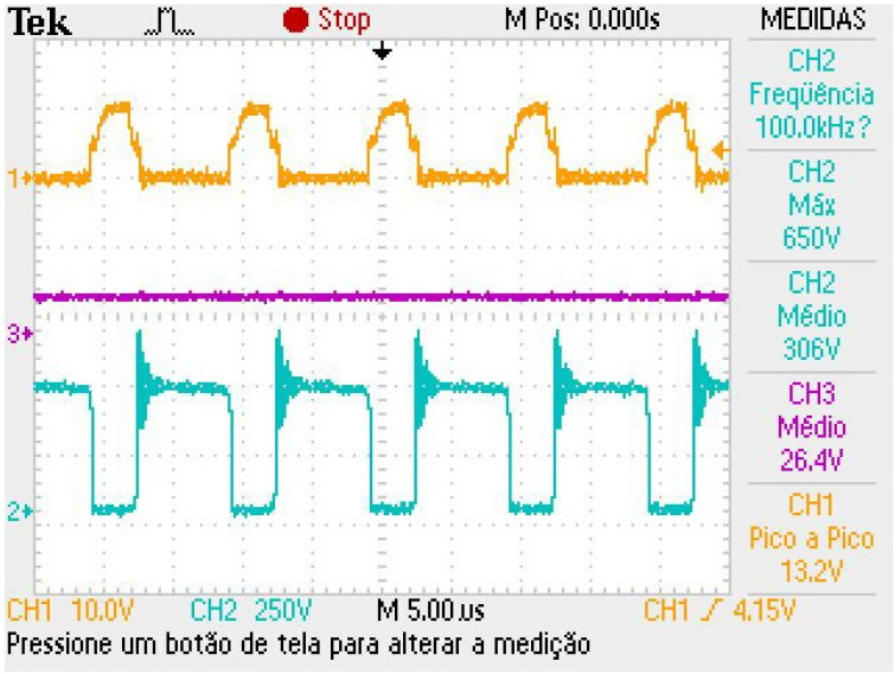
UC3844 – Flyback:
Tensão de Saída: 24V
Corrente Máxima: 2,5 A



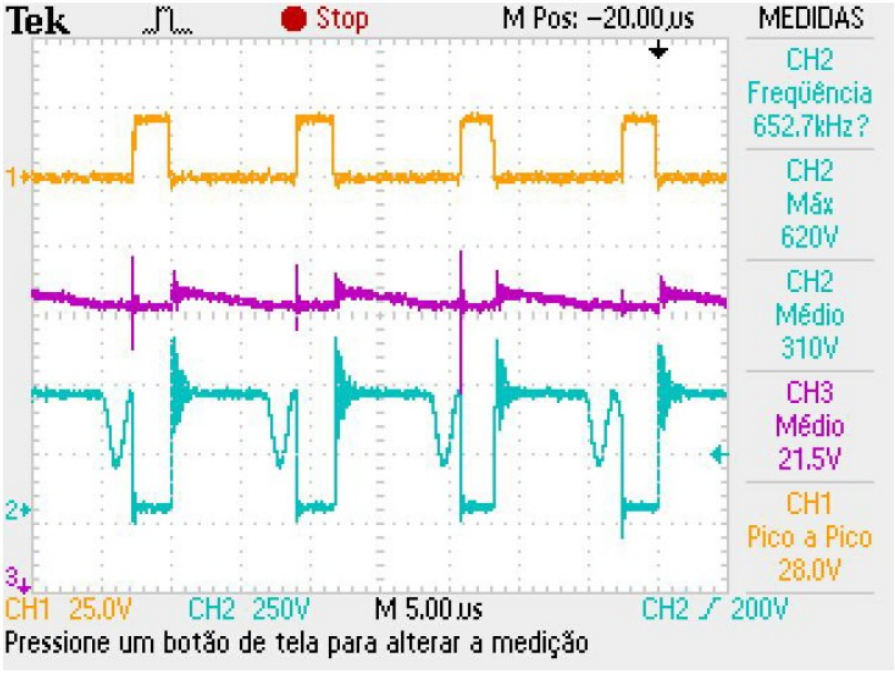
Fontes Chaveadas

Exemplo de uma fonte chaveada:

Semestre 2019-1: Bruna Martini e Rafael Reis



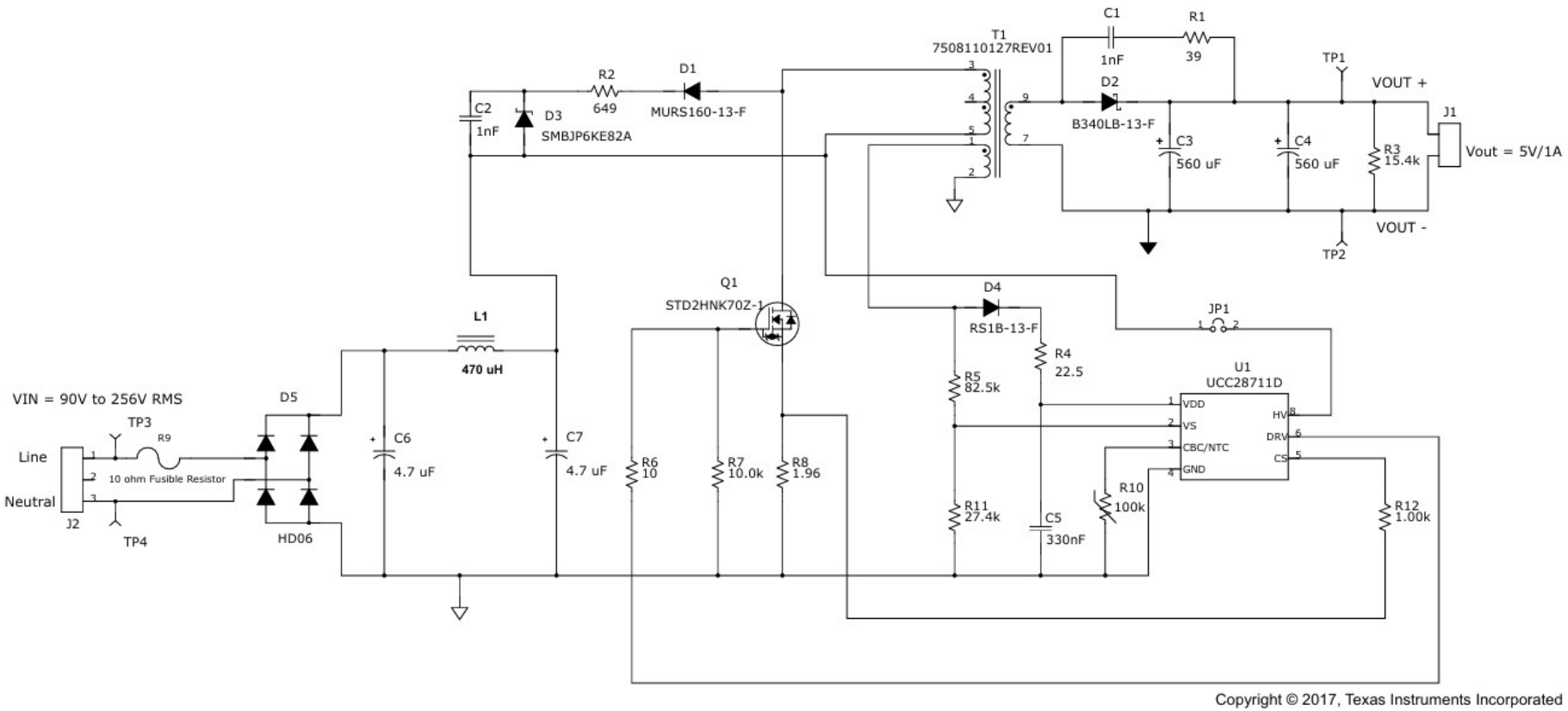
PWM: Gerador de Função



UC3844 em malha fechada

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Estágio de conversão CC-CC de uma fonte chaveada: UCC28712D



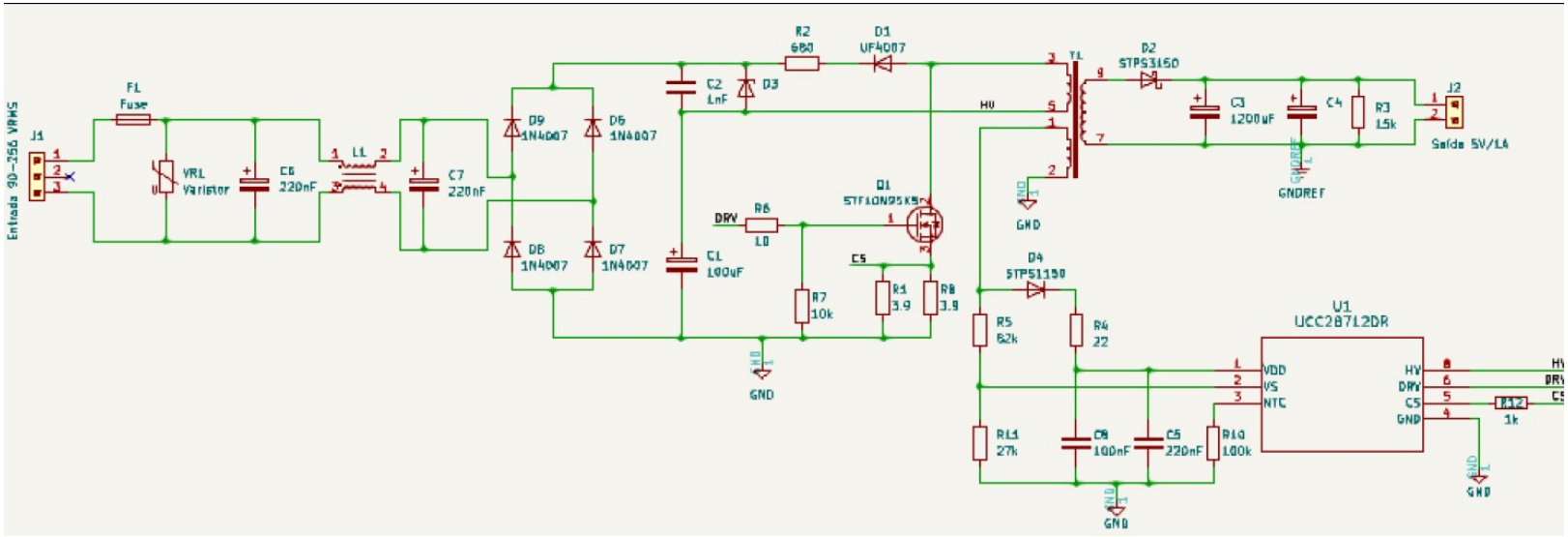
Copyright © 2017, Texas Instruments Incorporated

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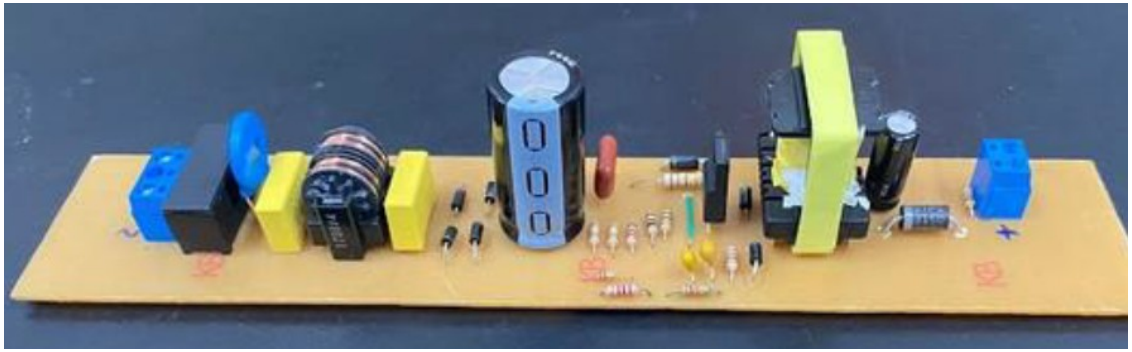
Exemplo de uma fonte chaveada: UCC28712D

Semestre 2025-2: Greicili dos Santos Ferreira e Jonathan C. C. Bonette

Diagrama Esquemático



UCC28712D – Flyback:
Tensão de Saída: 5 V
Corrente Máxima: 1 A



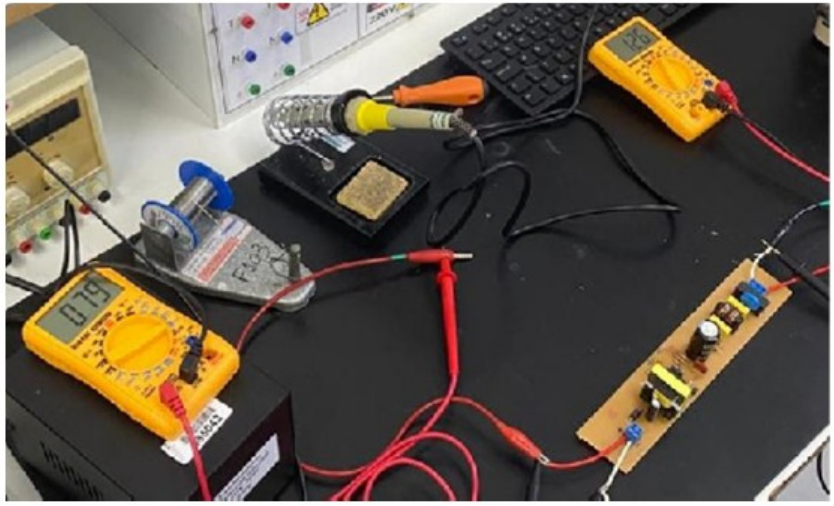
Fontes Chaveadas

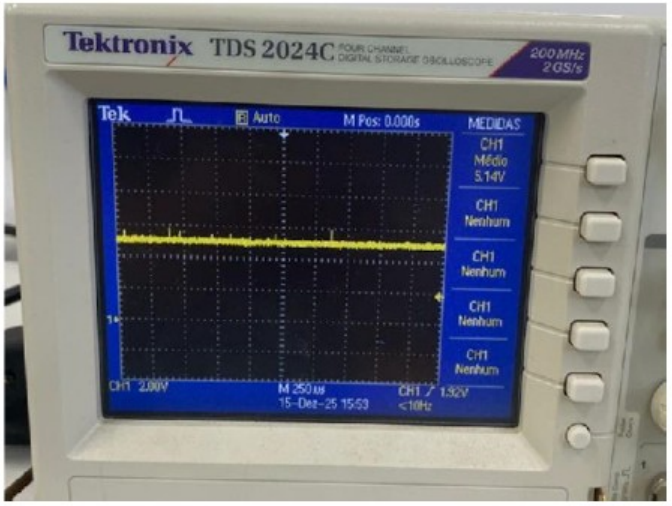
INSTITUTO FEDERAL
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Exemplo de uma fonte chaveada: UCC28712D

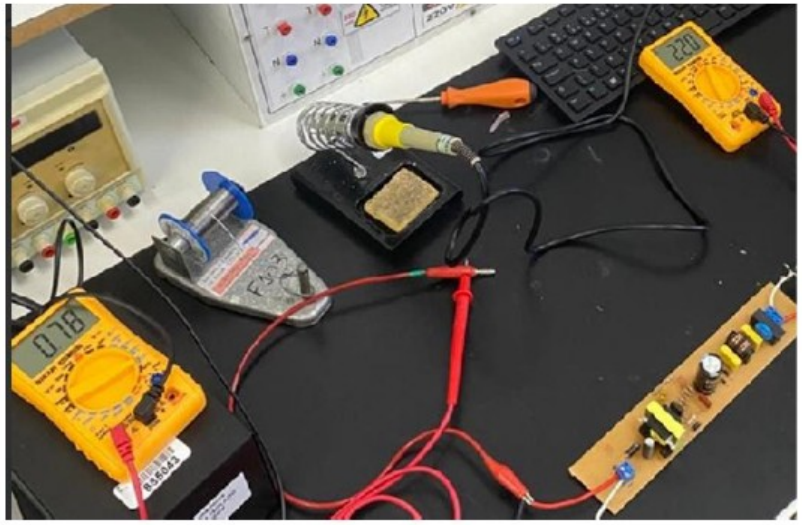
Semestre 2025-2: Greicili dos Santos Ferreira e Jonathan C. C. Bonette

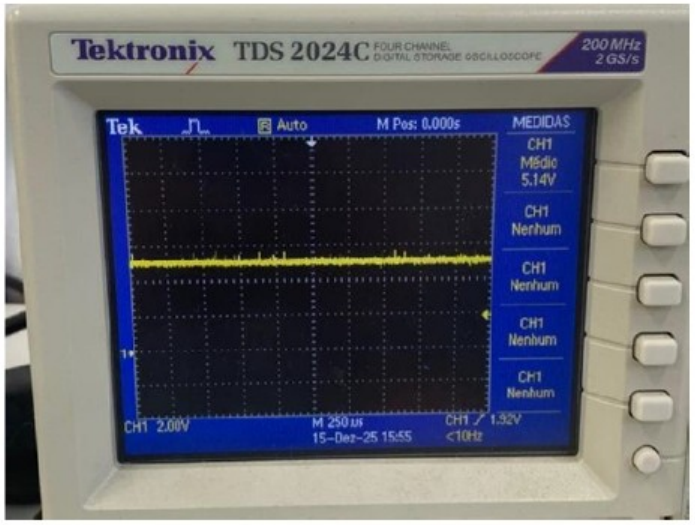
127 V





220 V



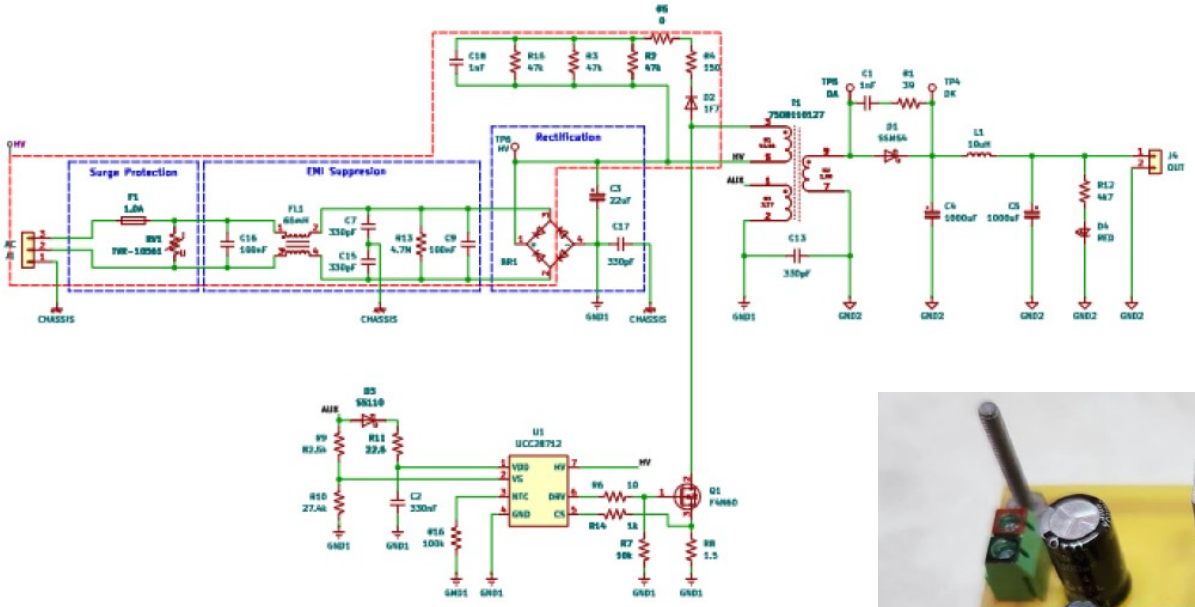


Fontes Chaveadas

Exemplo de uma fonte chaveada: UCC28712D

Semestre 2026-1: Nívea Luiza Cruz e Sofia Maia Lee

Diagrama Esquemático



UCC28712D – Flyback:
Tensão de Saída: 5 V
Corrente Máxima: 1 A

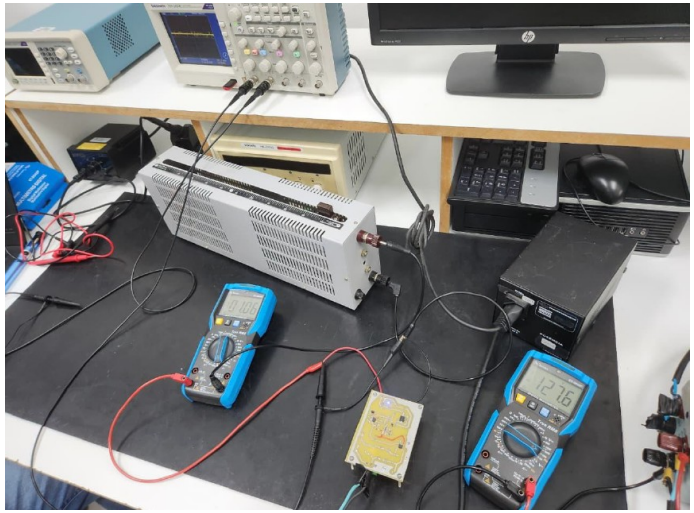


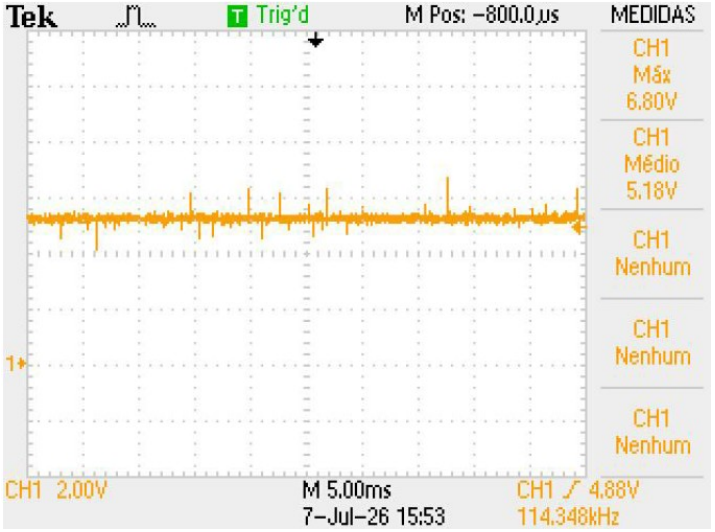
Fontes Chaveadas

Exemplo de uma fonte chaveada: UCC28712D

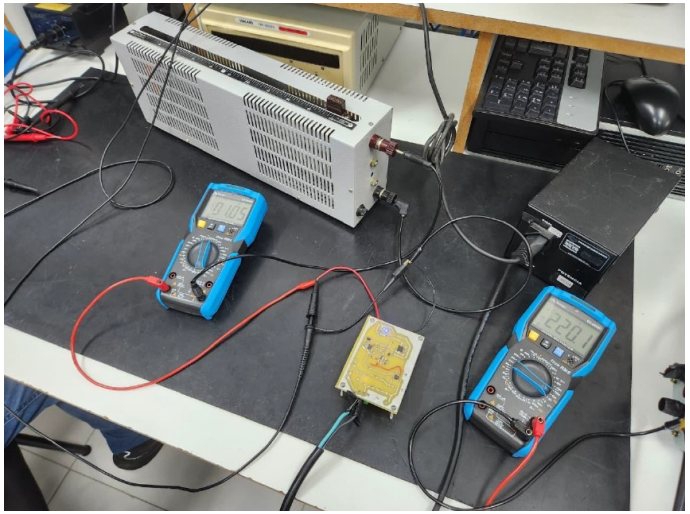
Semestre 2026-1: Nívea Luiza Cruz e Sofia Maia Lee

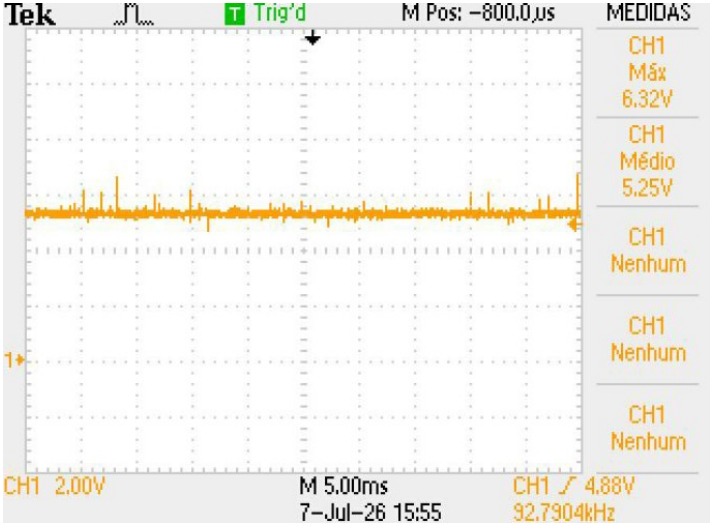
127 V





220 V



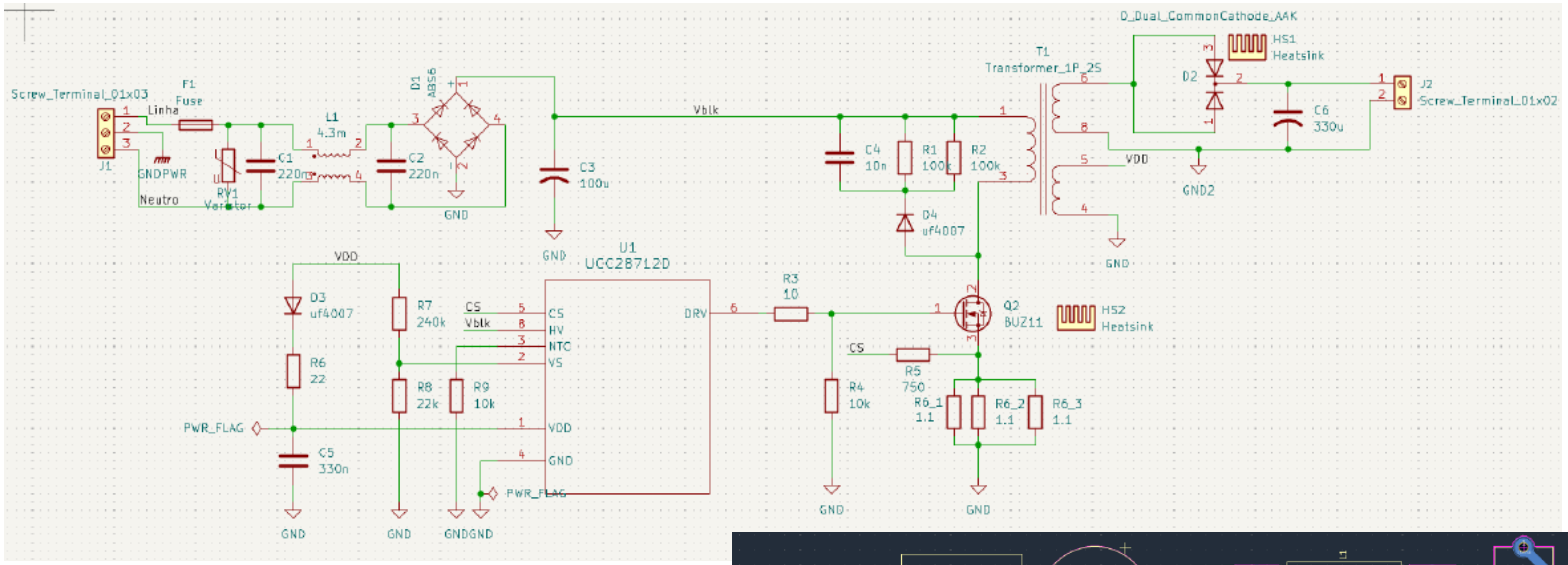


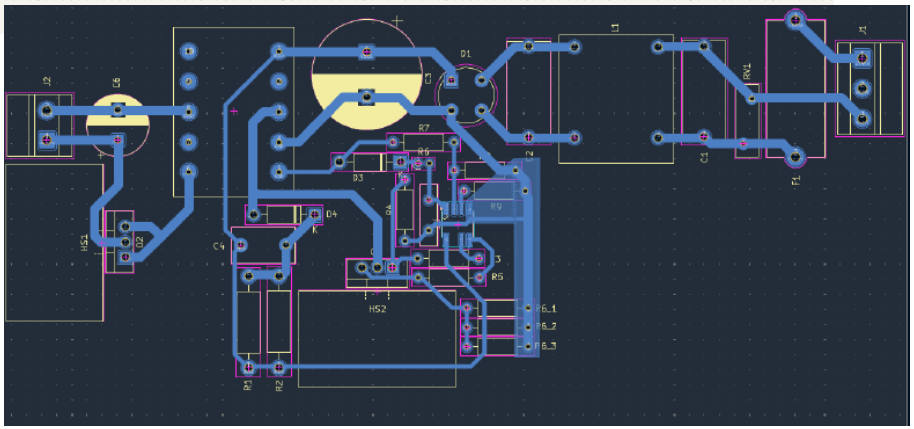
Fontes Chaveadas

Exemplo de uma fonte chaveada: UCC28712D

Semestre 2026-1: Gabriel L. Marodin, João Felipe C. Kloppel

Diagrama Esquemático





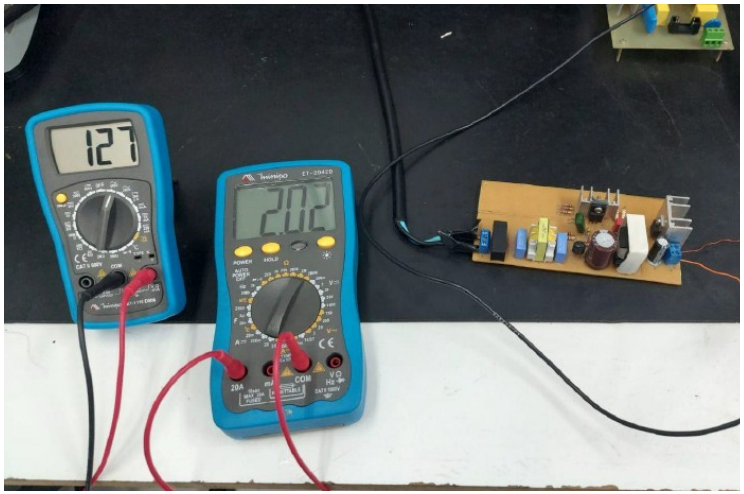
UCC28712D – Flyback:
Tensão de Saída: 12 V
Corrente Máxima: 3 A

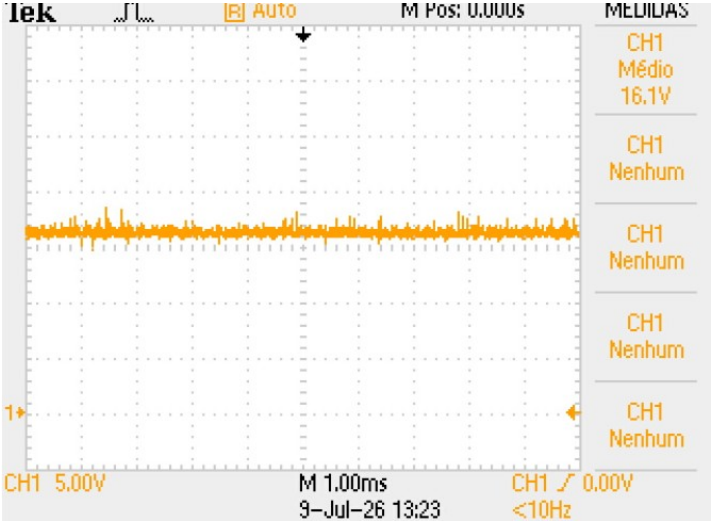
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Exemplo de uma fonte chaveada: UCC28712D

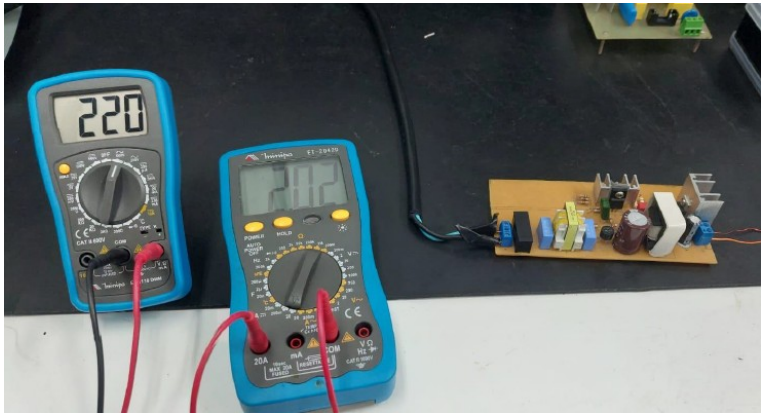
Semestre 2026-1: Gabriel L. Marodin, João Felipe C. Kloppel

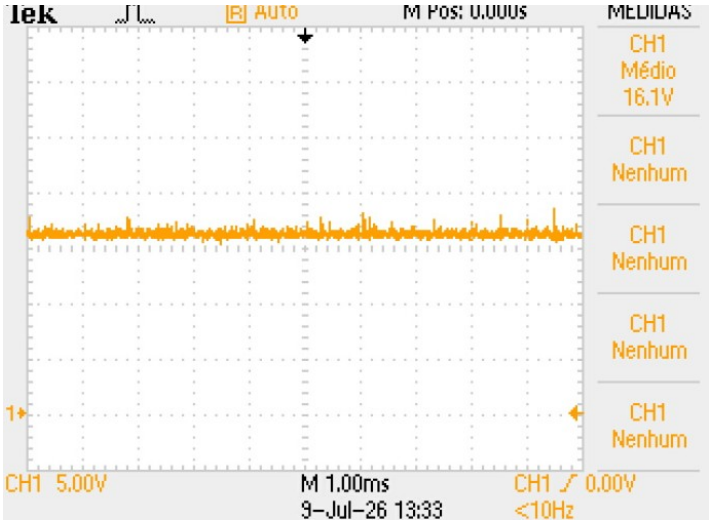
127 V





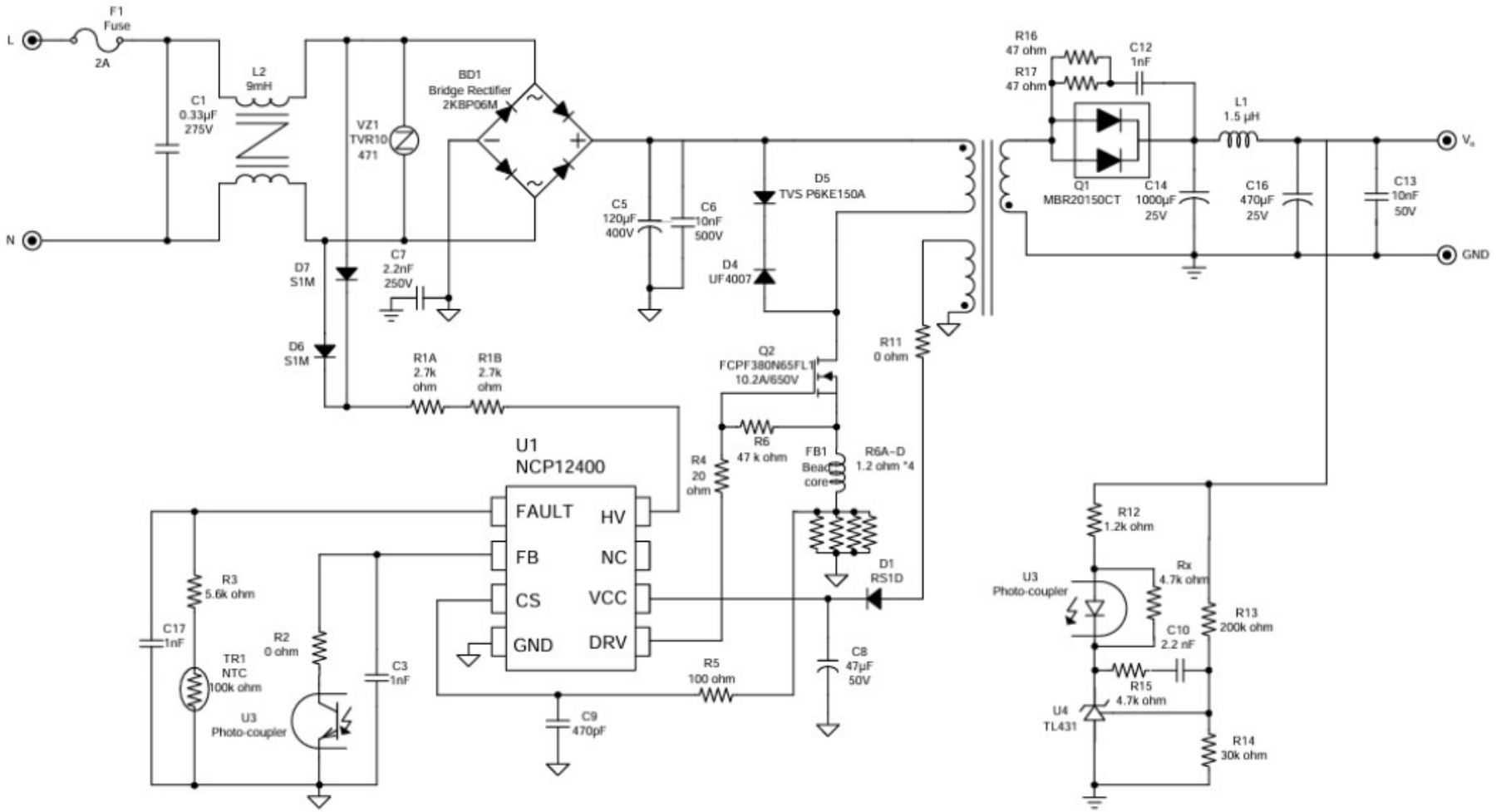
220 V





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Estágio de conversão CC-CC de uma fonte chaveada: NCP12400

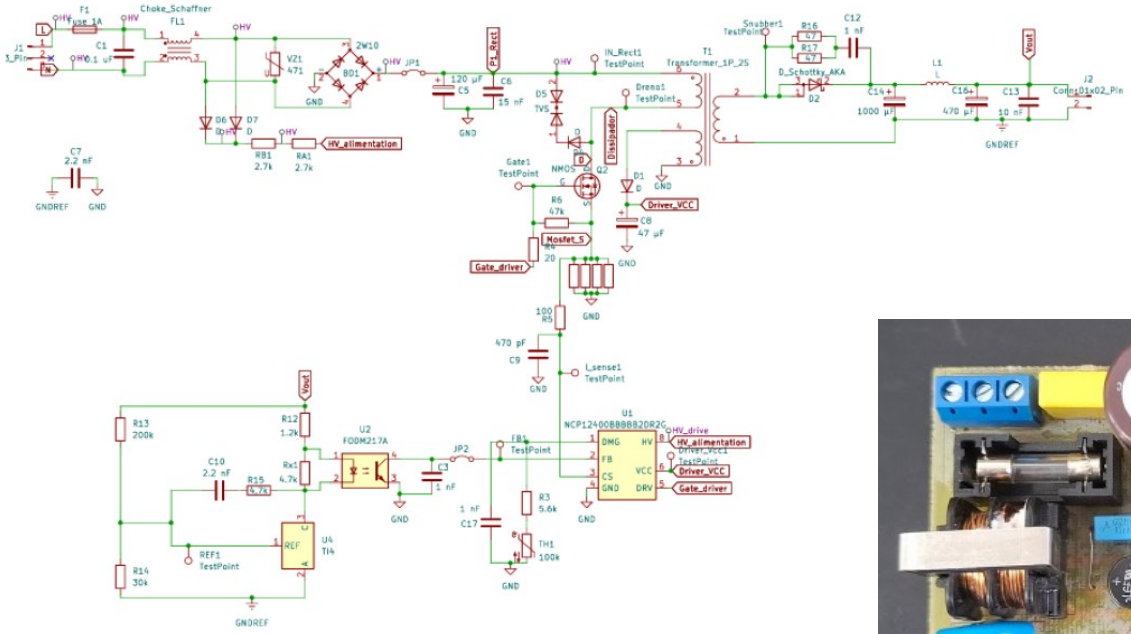


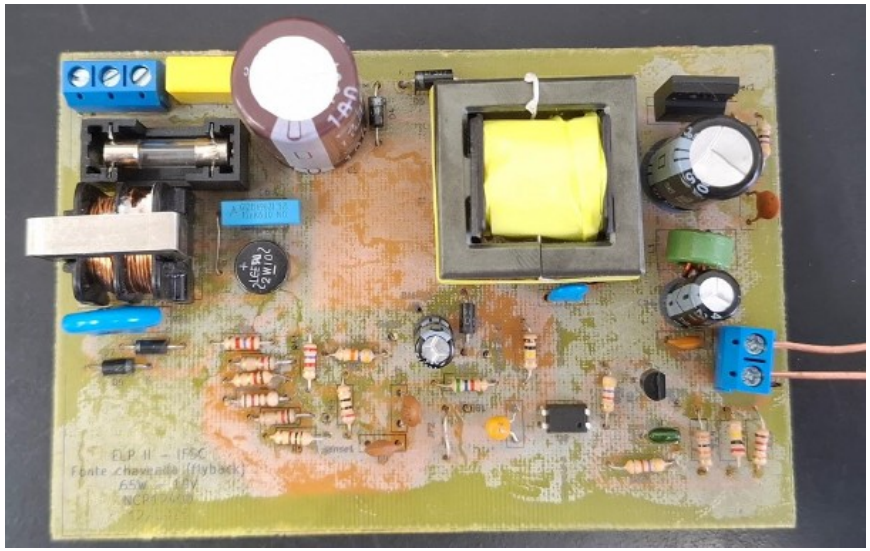
Fontes Chaveadas

Exemplo de uma fonte chaveada: NCP12400

Semestre 2025-2: Christopher Costa e Rafael Suzin

Diagrama Esquemático





NCP12400 – Flyback:
Tensão de Saída: 19,5 V
Corrente Máxima: 3,42 A
Potência = 65 W

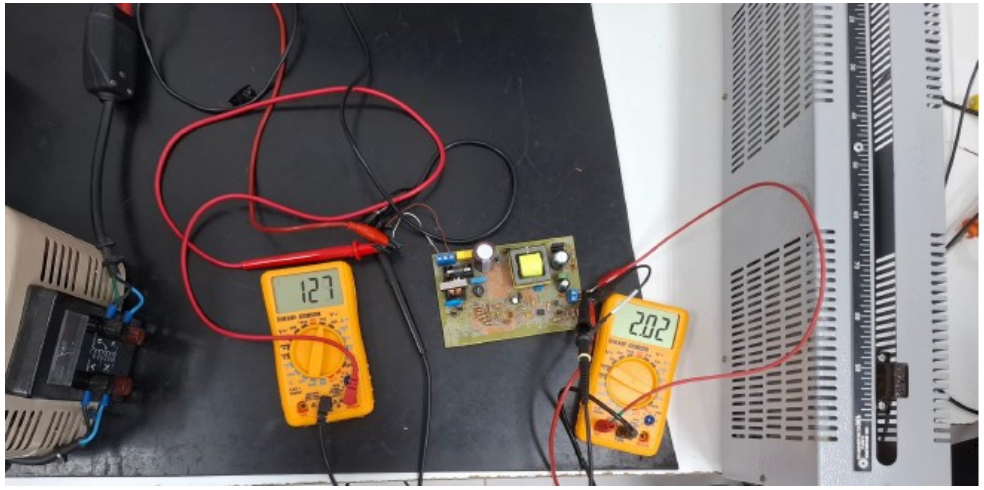
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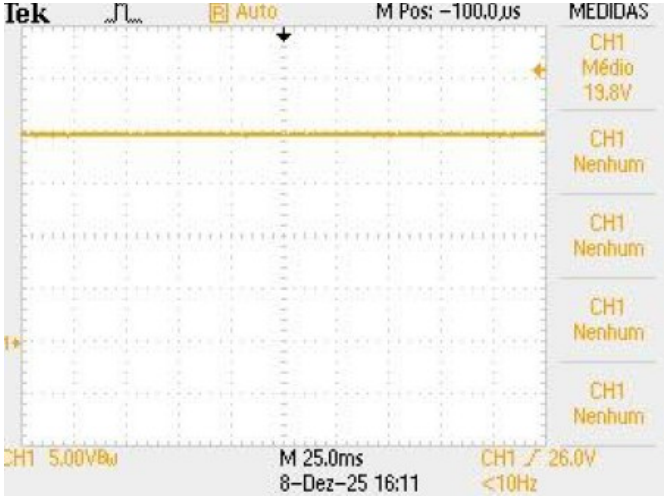
INSTITUTO FEDERAL
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Exemplo de uma fonte chaveada: UCC28712D

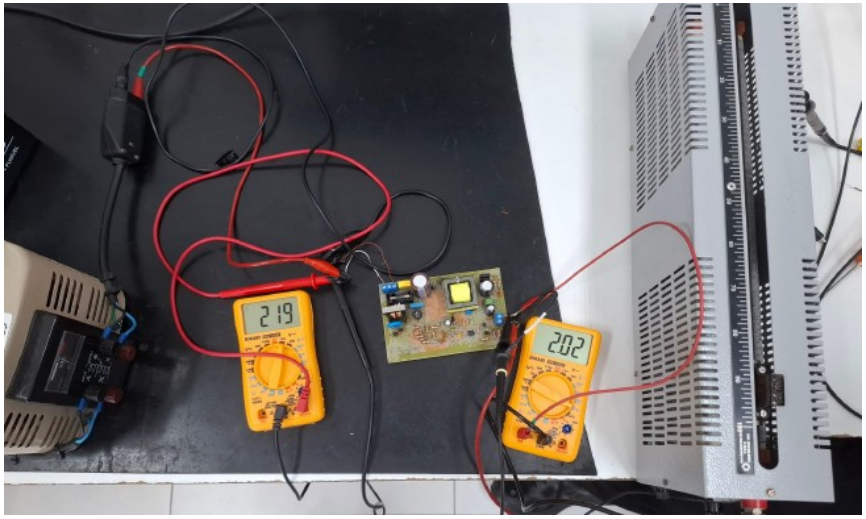
Semestre 2025-2: Christopher Costa e Rafael Suzin

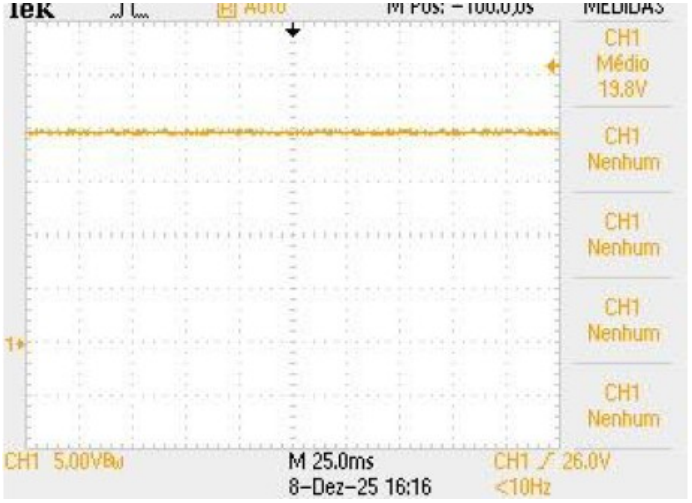
127 V



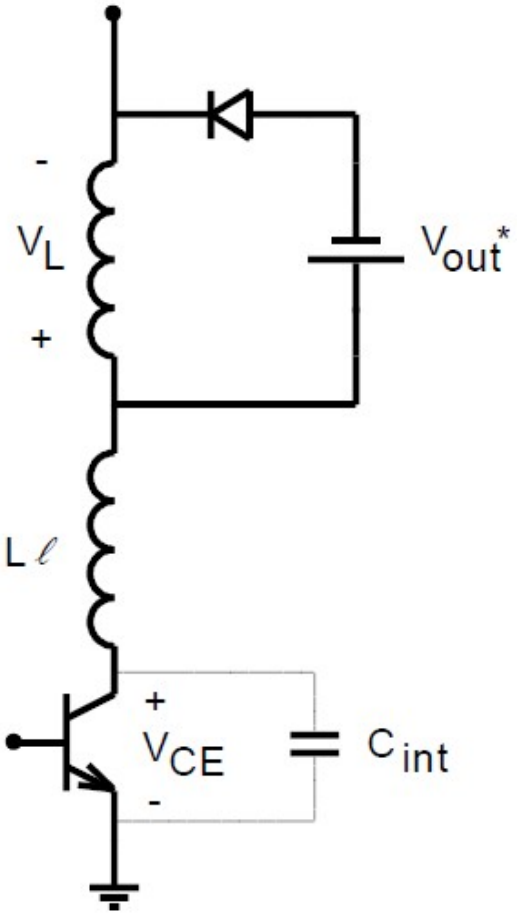


220 V

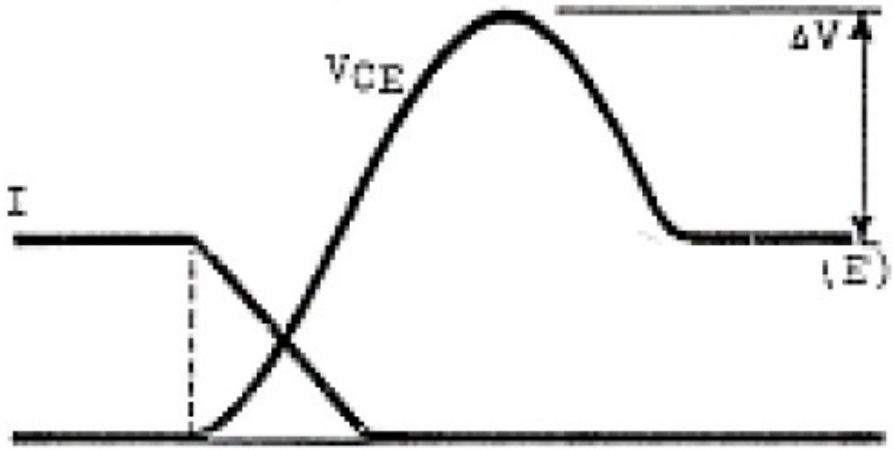




Grampeamento da tensão no interruptor

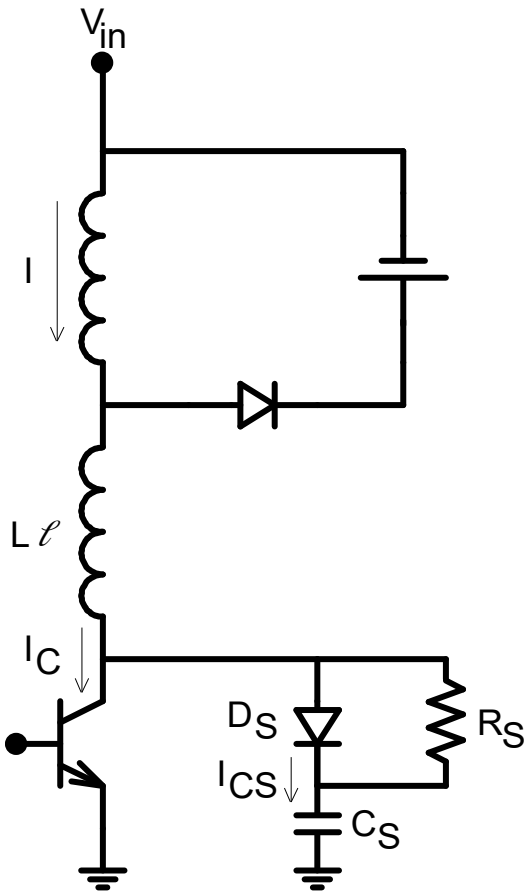


Conversor Flyback

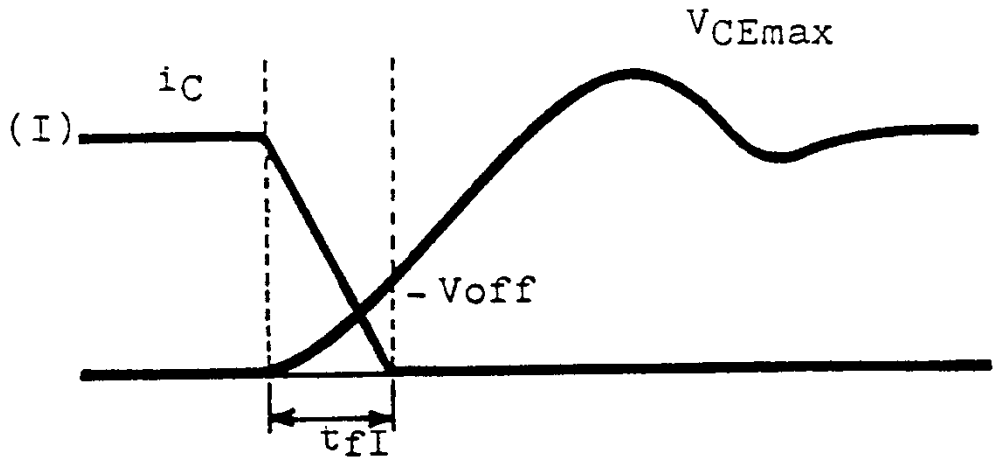


Sobretensão causada por Ll

Grampeamento da tensão no interruptor



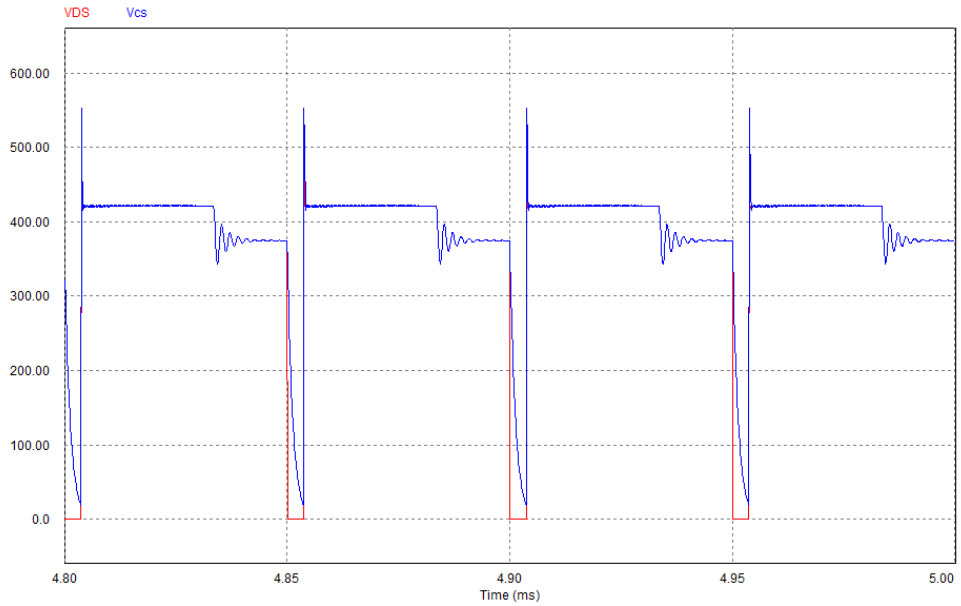
Emprego de snubber



Controle da sobretensão com o snubber

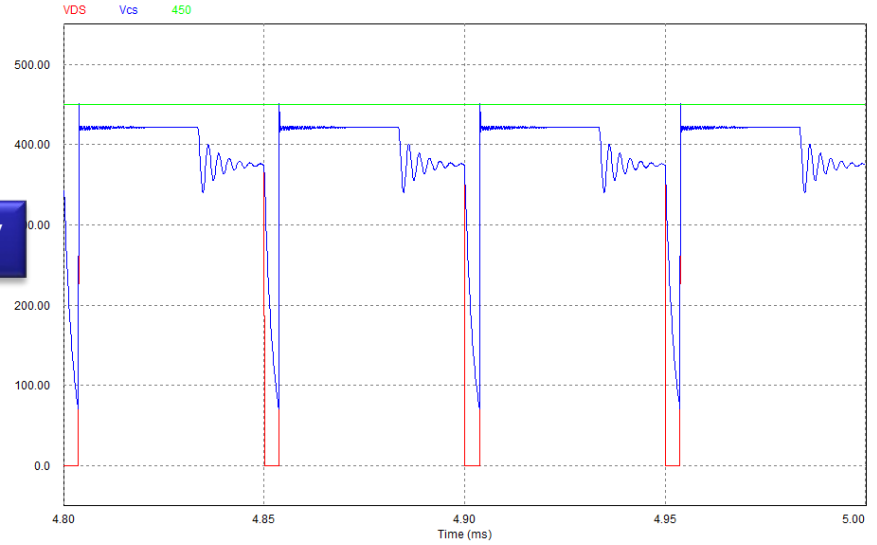
Grampeamento da tensão no interruptor

Com o uso do Circuito Snuber com valores diferentes dos componentes



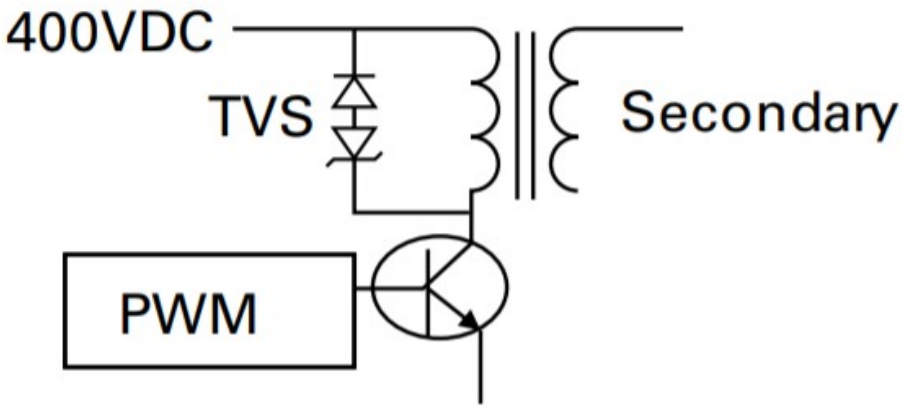
Sobretensão de 553 V

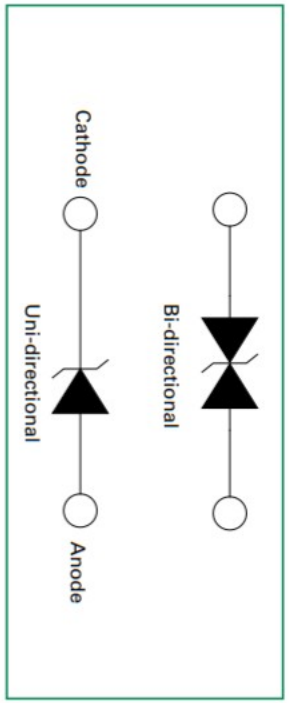
Sobretensão de 450 V

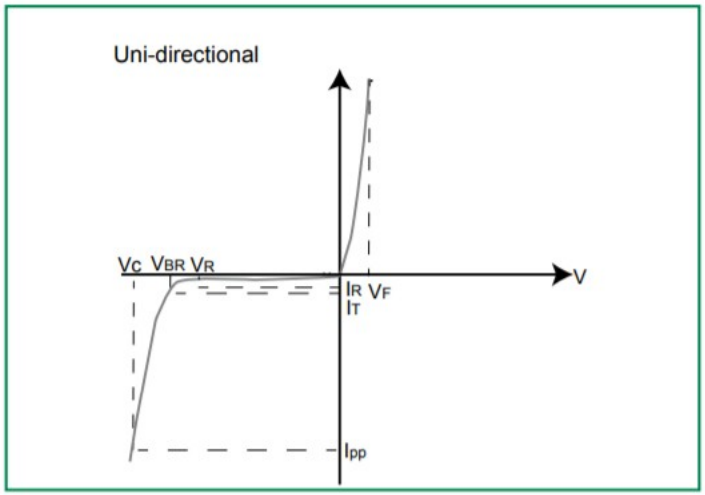


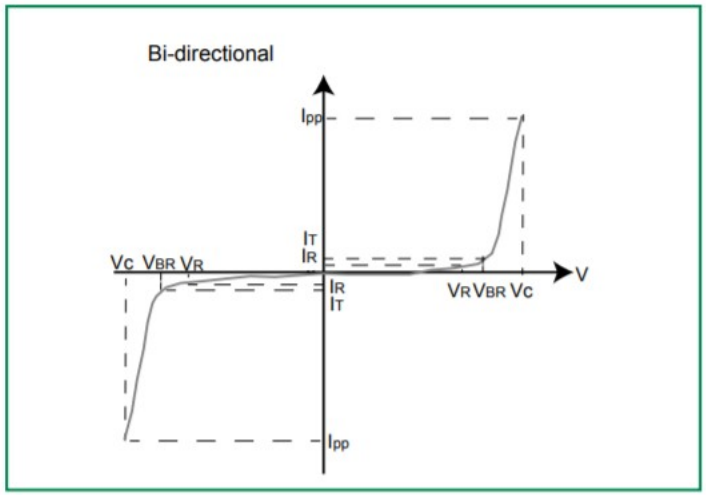
Grampeamento da tensão no interruptor

TVS (Transient Voltage Supression) ou TRANSIL







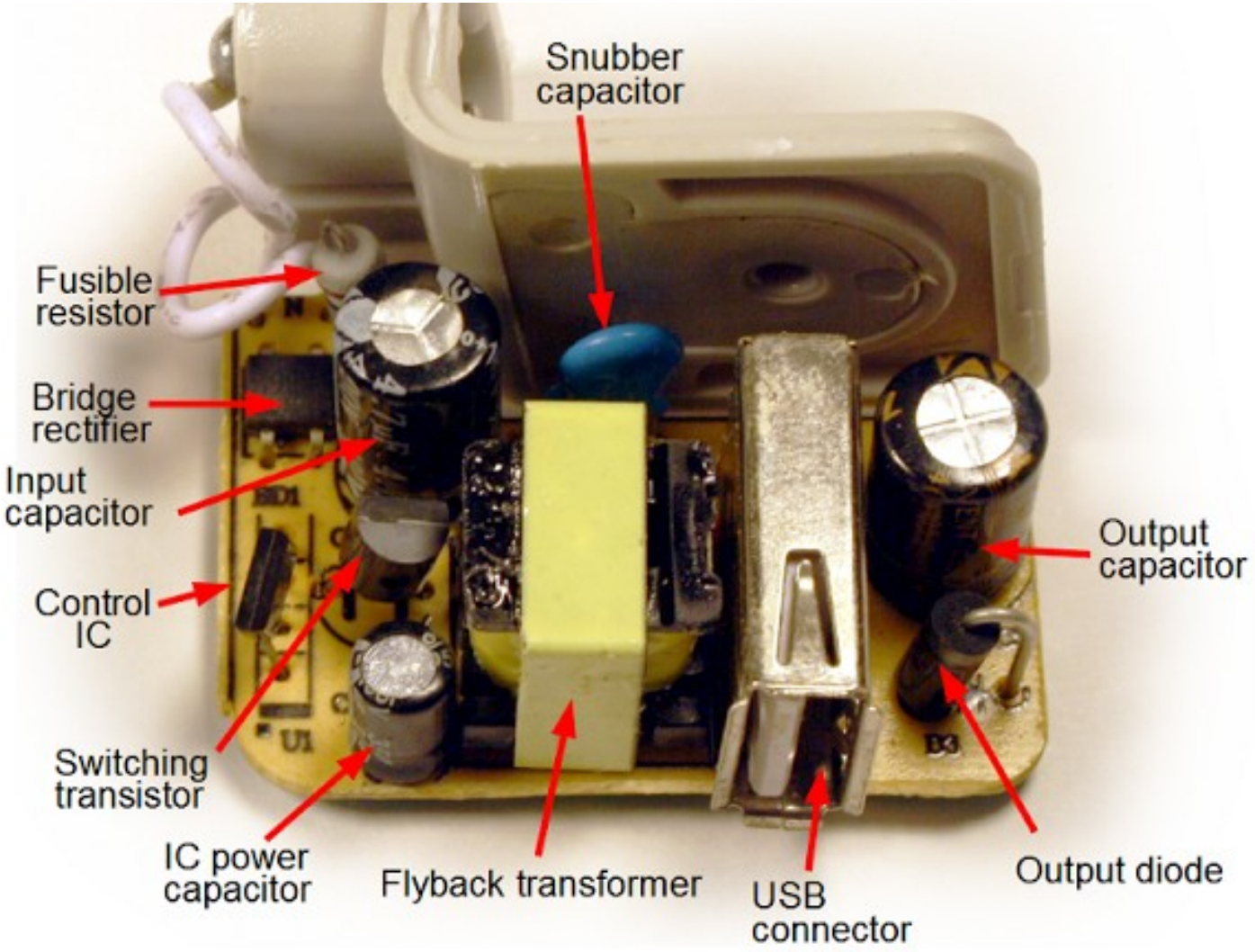


Fontes Chaveadas



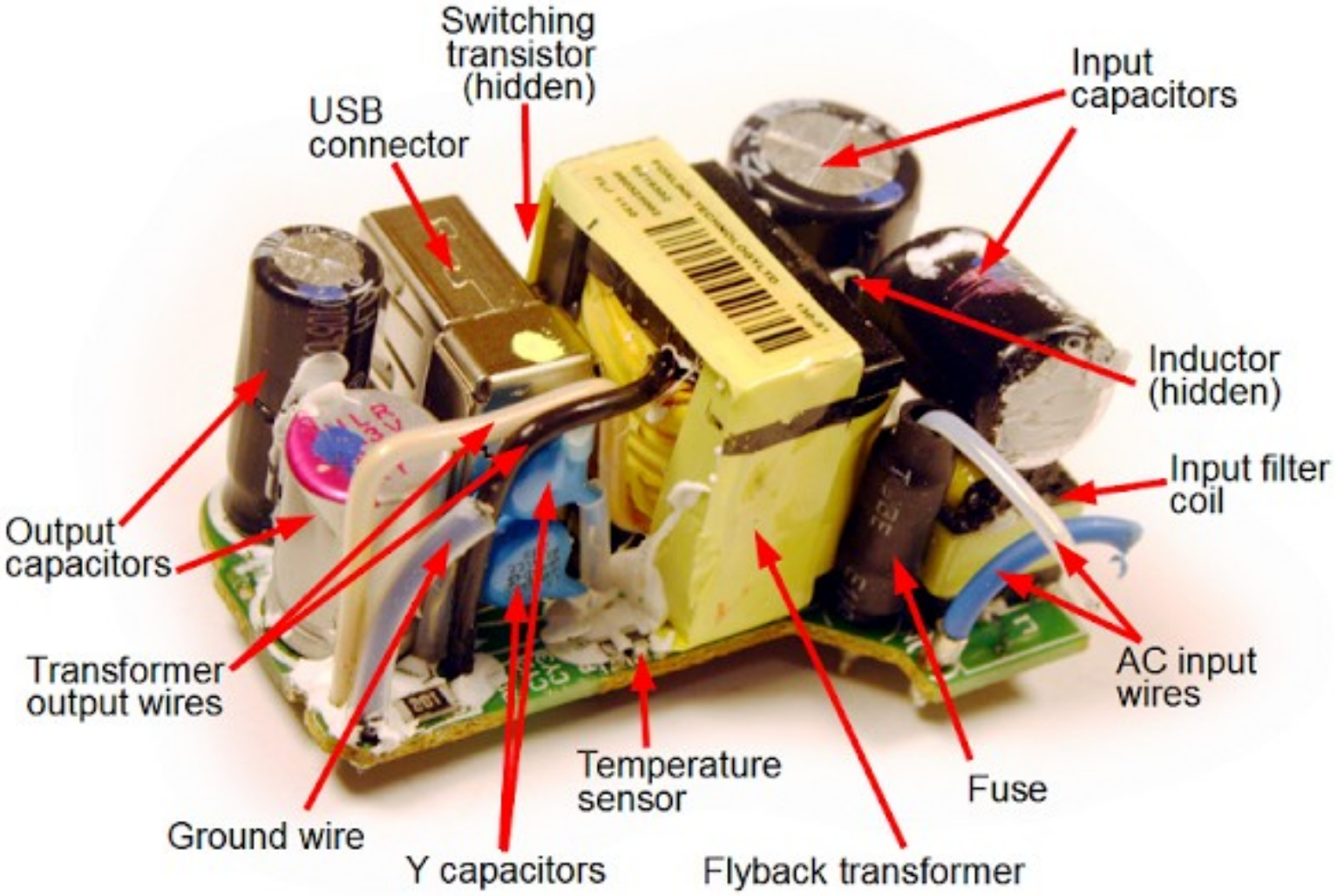
http://www.righto.com/2014/05/a-look-inside-ipad-chargers-pricey.html

Fontes Chaveadas



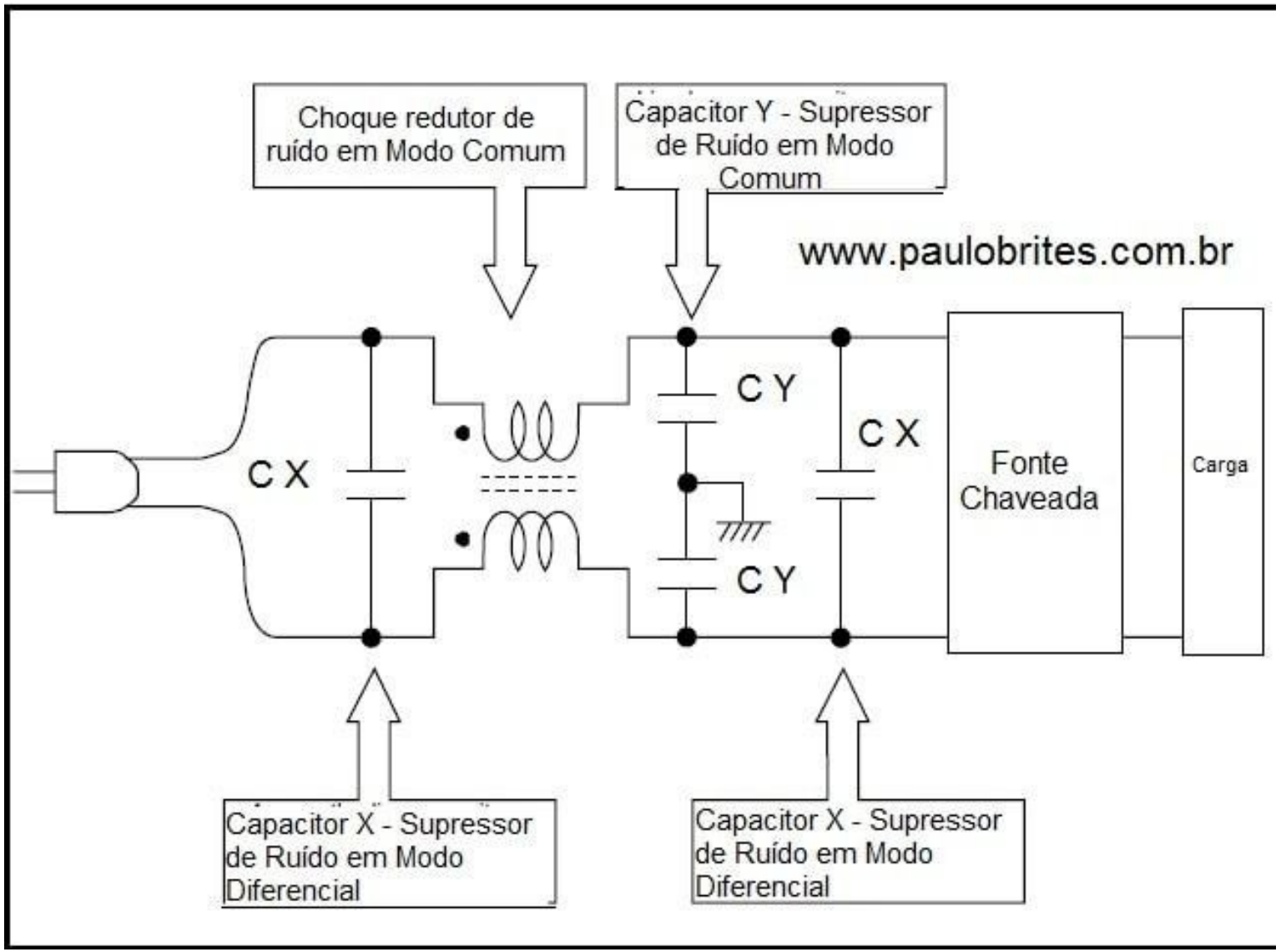
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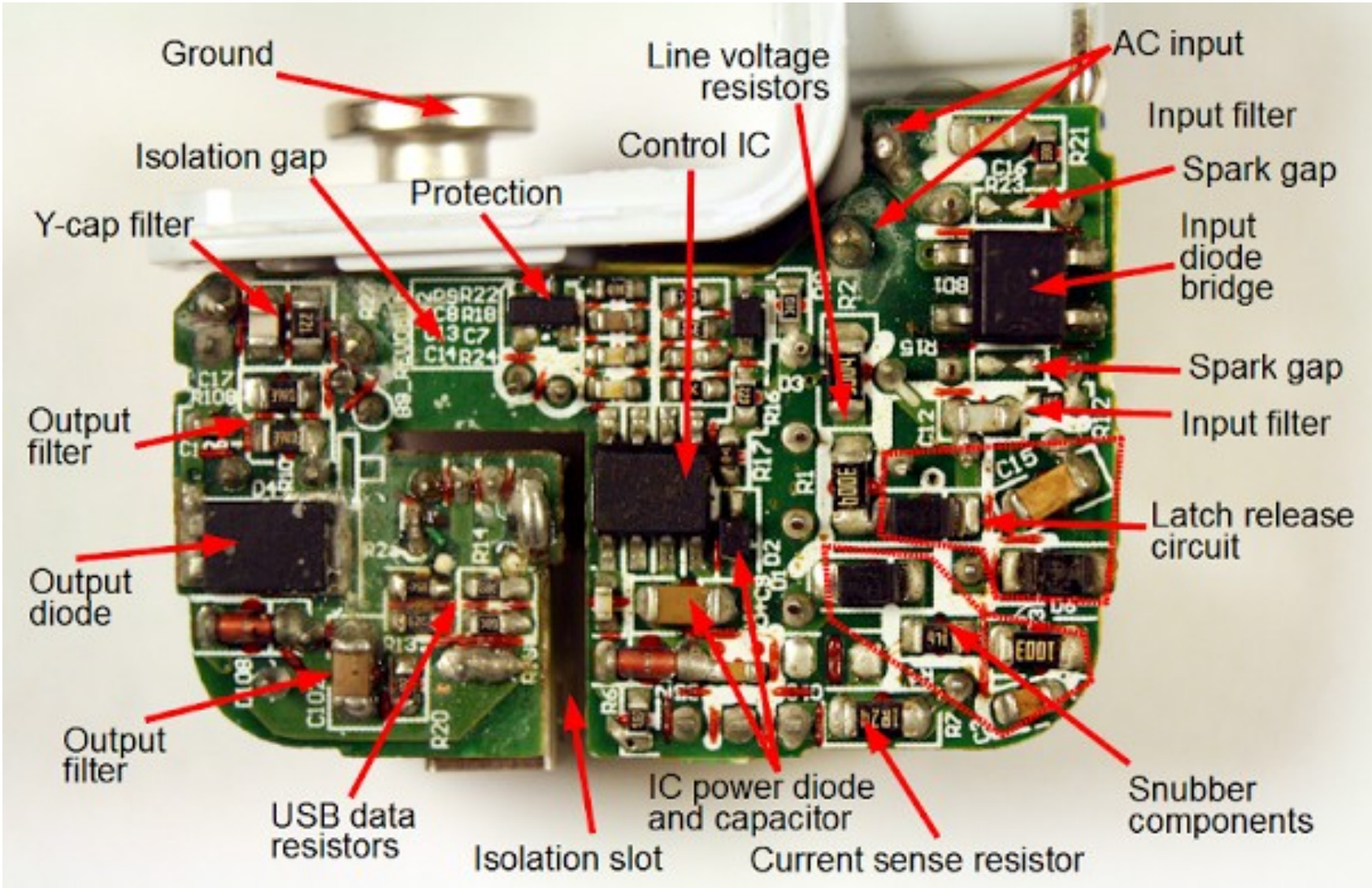


http://www.righto.com/2014/05/a-look-inside-ipad-chargers-pricey.html

Fontes Chaveadas

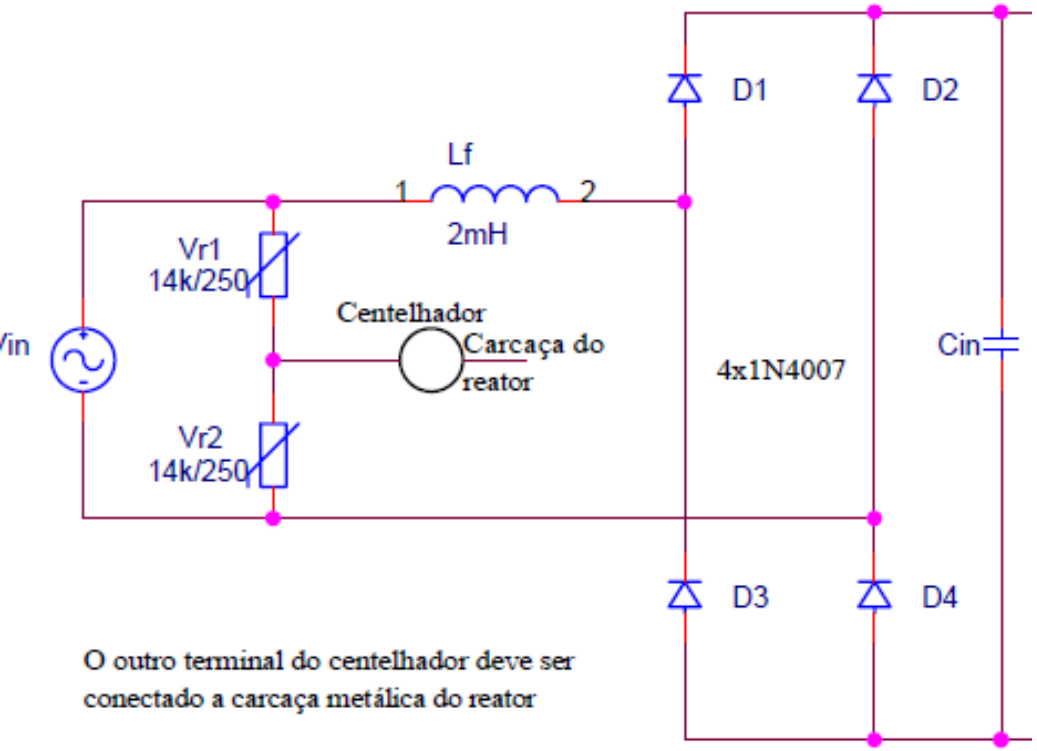


Fontes Chaveadas



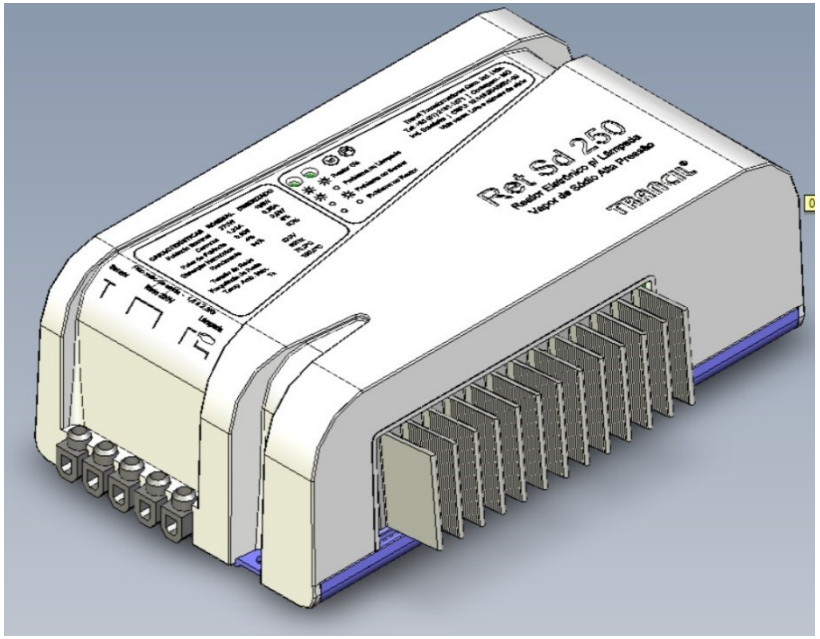
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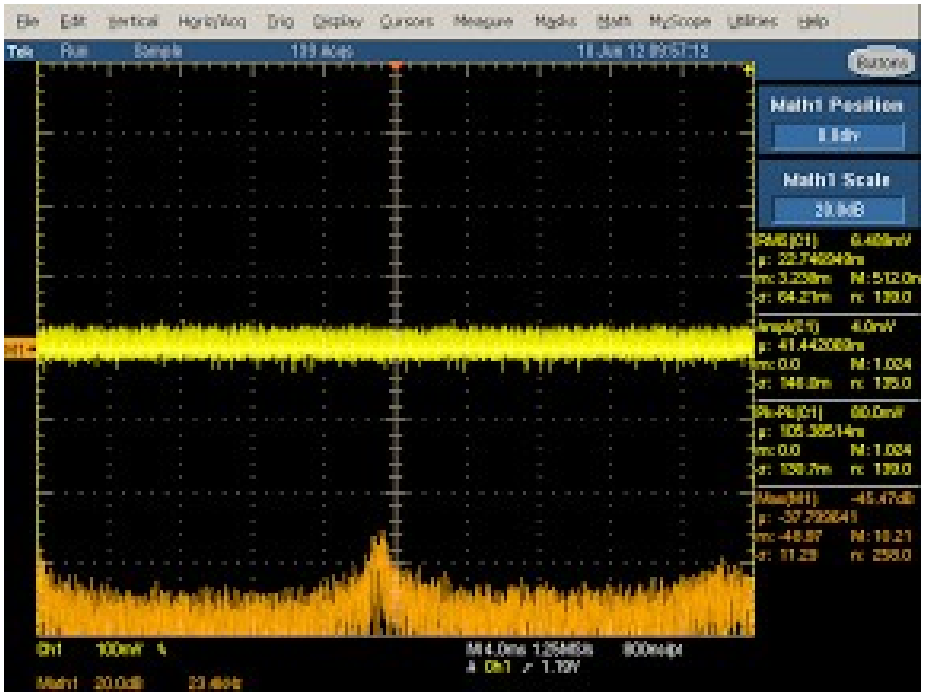


O outro terminal do centelhador deve ser conectado a carcaça metálica do reator

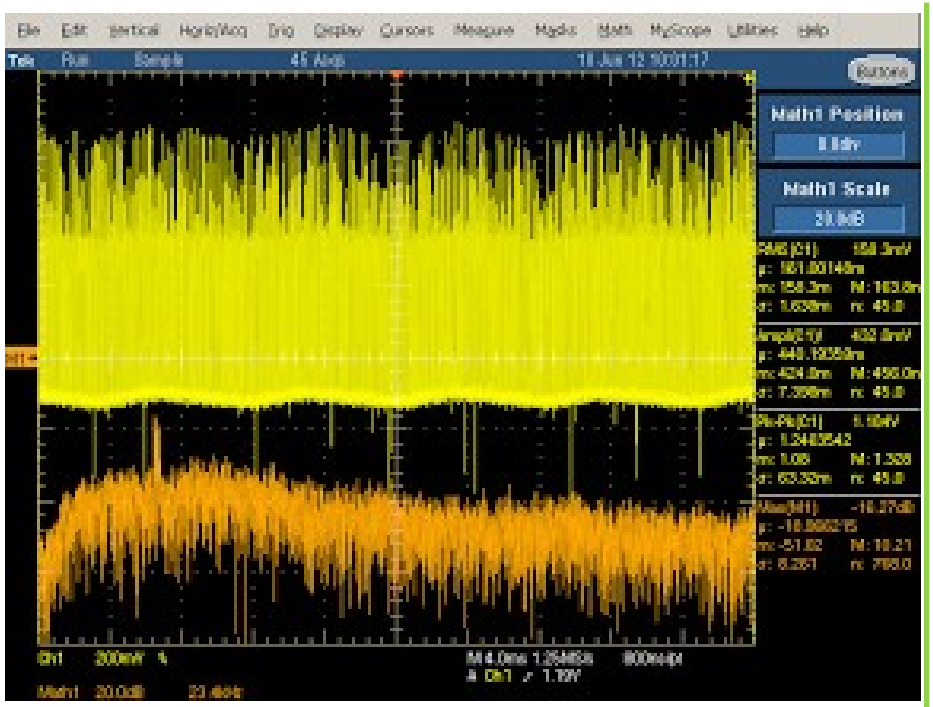




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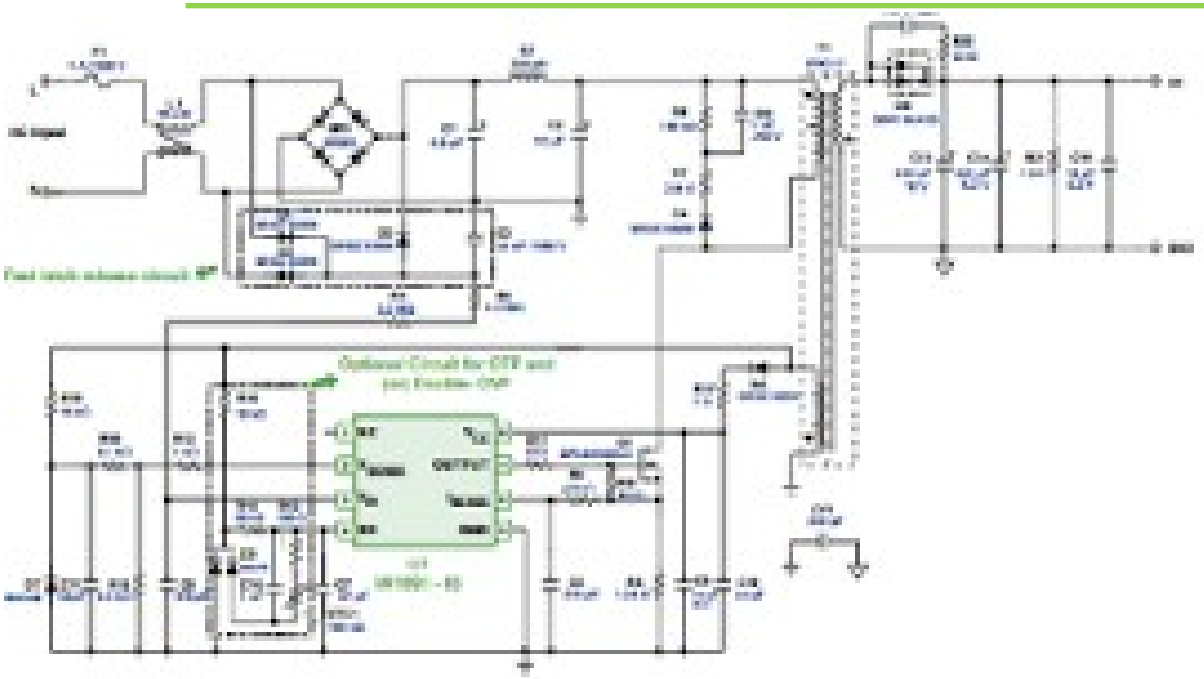
Carregador Original



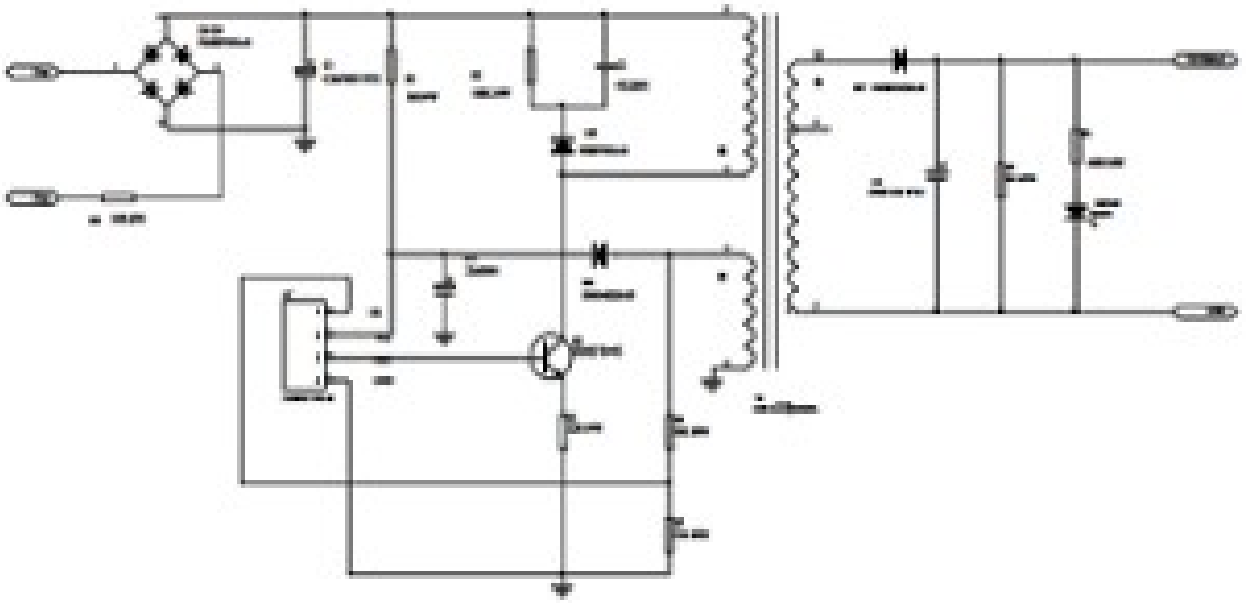
Carregador Alternativo

http://www.righto.com/2014/05/a-look-inside-ipad-chargers-pricey.html

Fontes Chaveadas

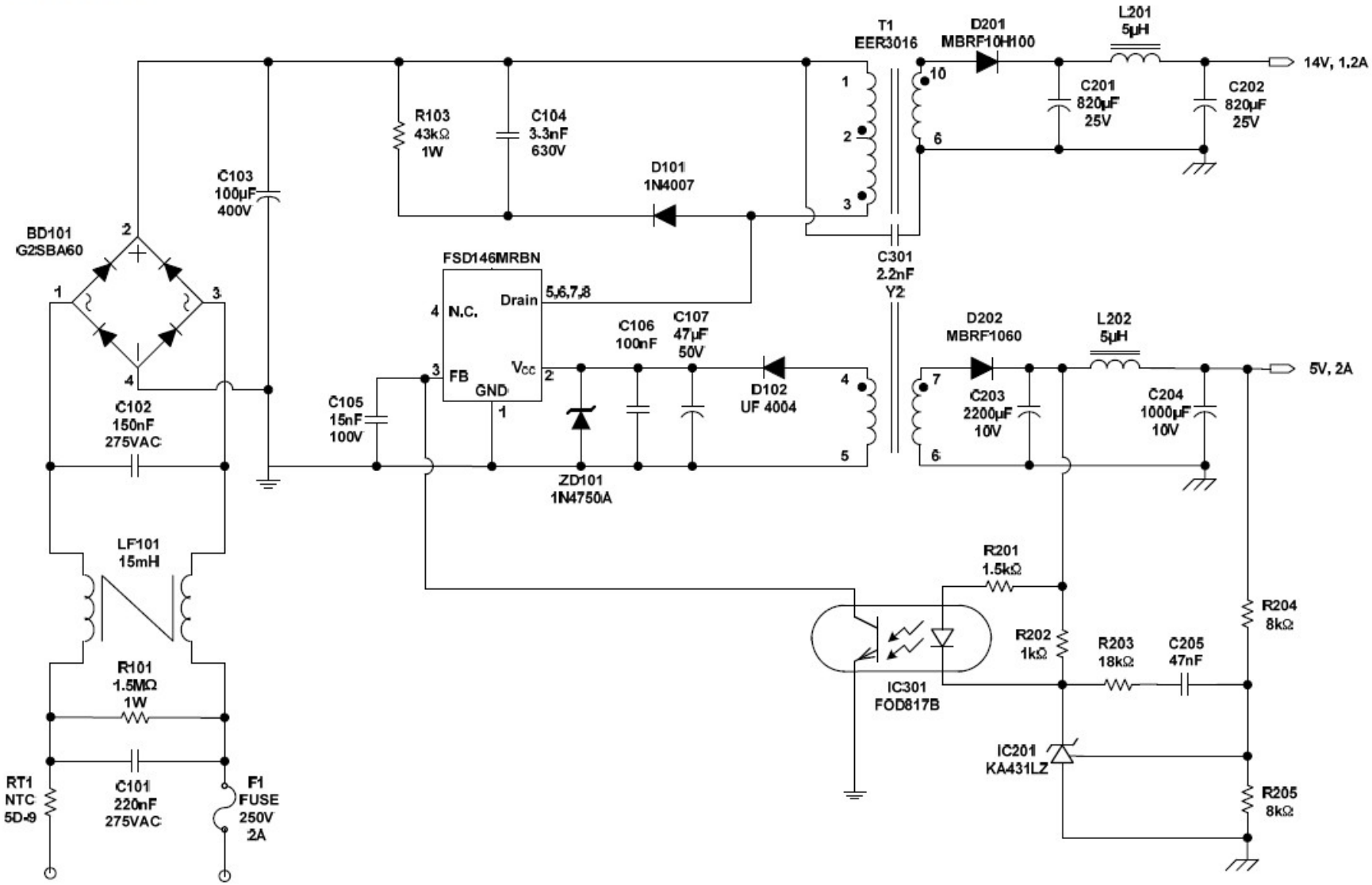


Esquemático
Carregador Original



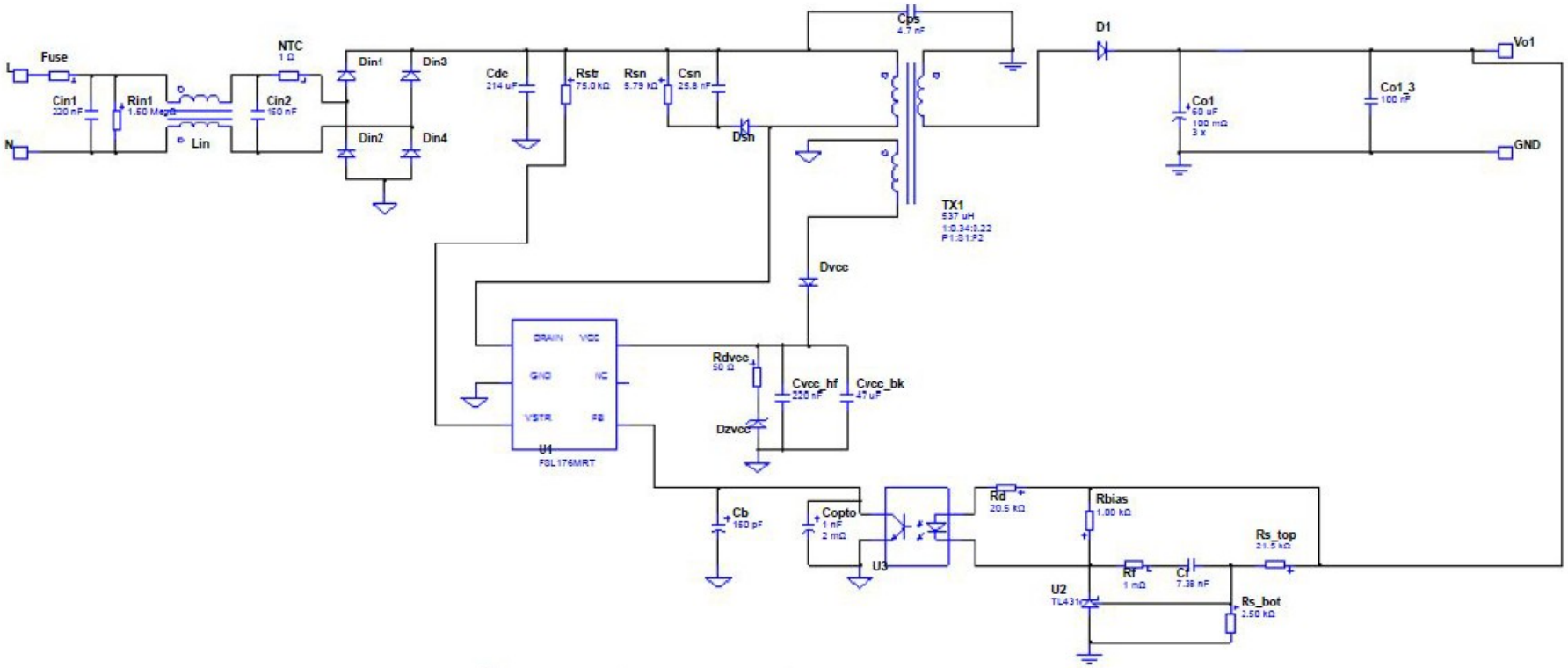
Esquemático
Carregador Alternativo

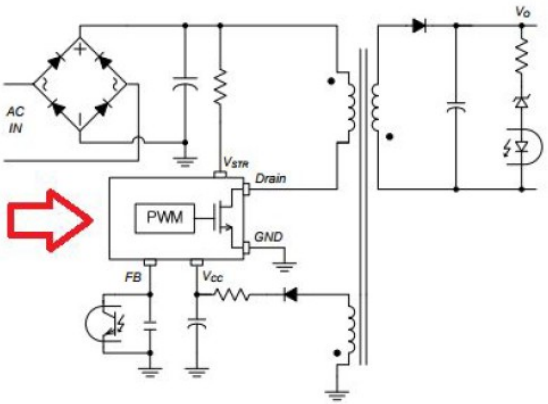
Fontes Chaveadas

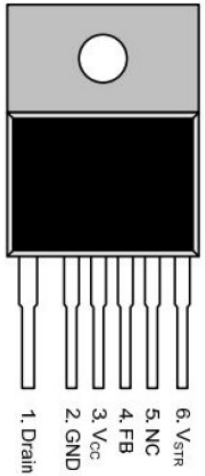


FSD146MRBN – Fonte para monitor de LCD

Fontes Chaveadas

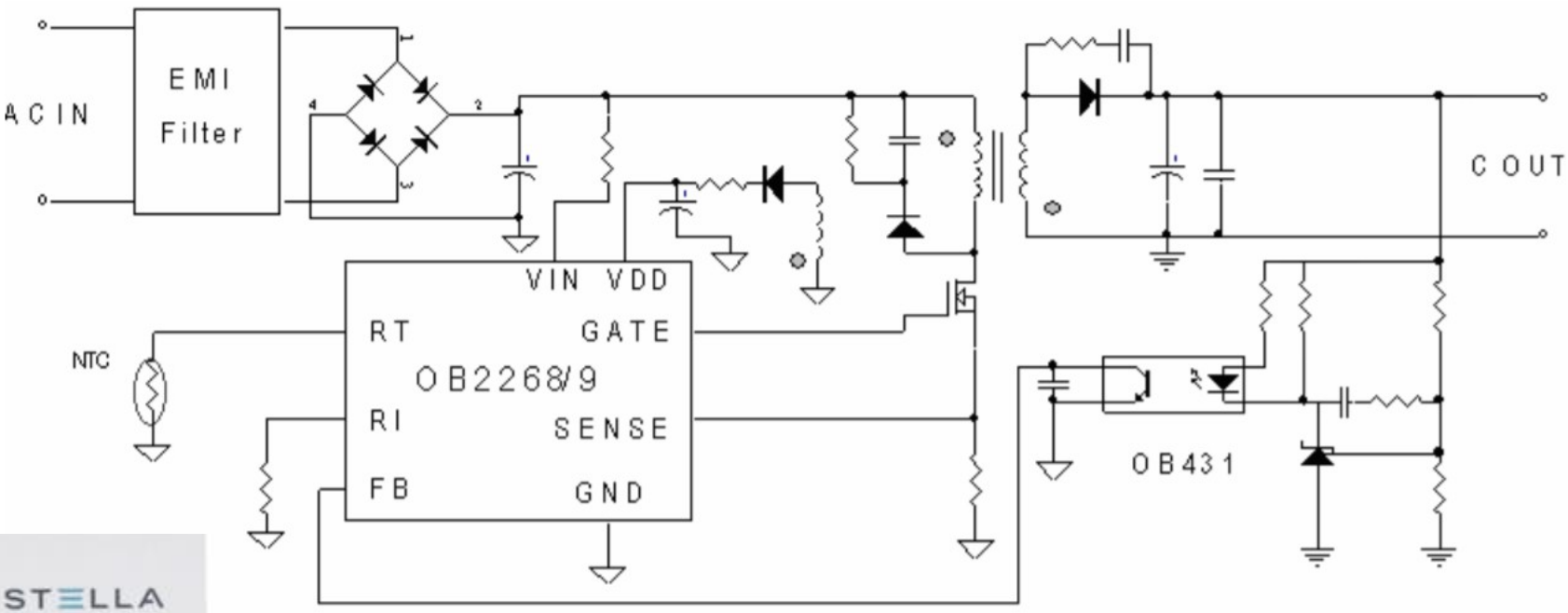






FSL176 – Carregador de Bateria

Fontes Chaveadas





Driver para Led 25 W

Fontes Chaveadas

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Fontes Alimentação para IIOT – Industrial IOT e Indústria 4.0

Features

Regulated Converter

- Ultra-wide input range 85-528VAC
- OVC III input rating without additional fuses
- Operating temperature range: -40°C to +80°C
- Overvoltage and overcurrent protected
- Class II installations (without FG)
- EMC compliant without external components
- No load power consumption <0.5W

Description

The RAC05-K/480 series of 5 watt AC/DC units are specially designed for harsh industrial and outdoor mains conditions. These PCB-mount power supplies are rated to OVC III conditions from 100-480VAC nominal input lines with phase-to-phase or single phase operation without any external components needed. The modules support an operating temperature range from -40°C to +80°C and come with fully protected outputs as well as EMC Class B compliance. All these features make them an ideal fit for integration into smart grid, renewable energy, smart metering and IoT applications.

Selection Guide

Part Number	Input Voltage Range [VAC]	Output Voltage [VDC]	Output Current [mA]	Efficiency typ (1) [%]	Max. Capacitive Load (2) [µF]
RAC05-05SK/480	85-528	5	1000	63	10000
RAC05-12SK/480	85-528	12	420	65	1200

RECOM POWER

Fontes de alimentação para equipamentos industriais e de automação de fábrica são especiais. Devem operar em um ambiente considerado perigoso, mas também devem ser pequenos, eficientes, confiáveis e econômicos. Eles são uma combinação difícil de metas de design para alcançar.



Fontes Chaveadas

Fontes Alimentação para IIOT – Industrial IOT e Indústria 4.0

- Design compacto: tamanho pequeno, perfil baixo;
- Baixo consumo em Standby;
- Bom rendimento;
- Ampla faixa de tensão de alimentação (85 V até 528 V, CA);
- Ampla faixa de temperatura (−40 a + 85 C);
- Tensões CC de saída: 3,3, 5, 12, 15, 24 e 48 V;
- Proteções contra sobretensão, sobrecarga e curto-circuito;
- Certificações: EUA, Europeia e Internacional;
- Filtro de EMC;
- Baixo Custo;
- Três Anos de Garantia;
- Bom custo benefício;





Custo: Mouser
U$18,56 => 5V/1A

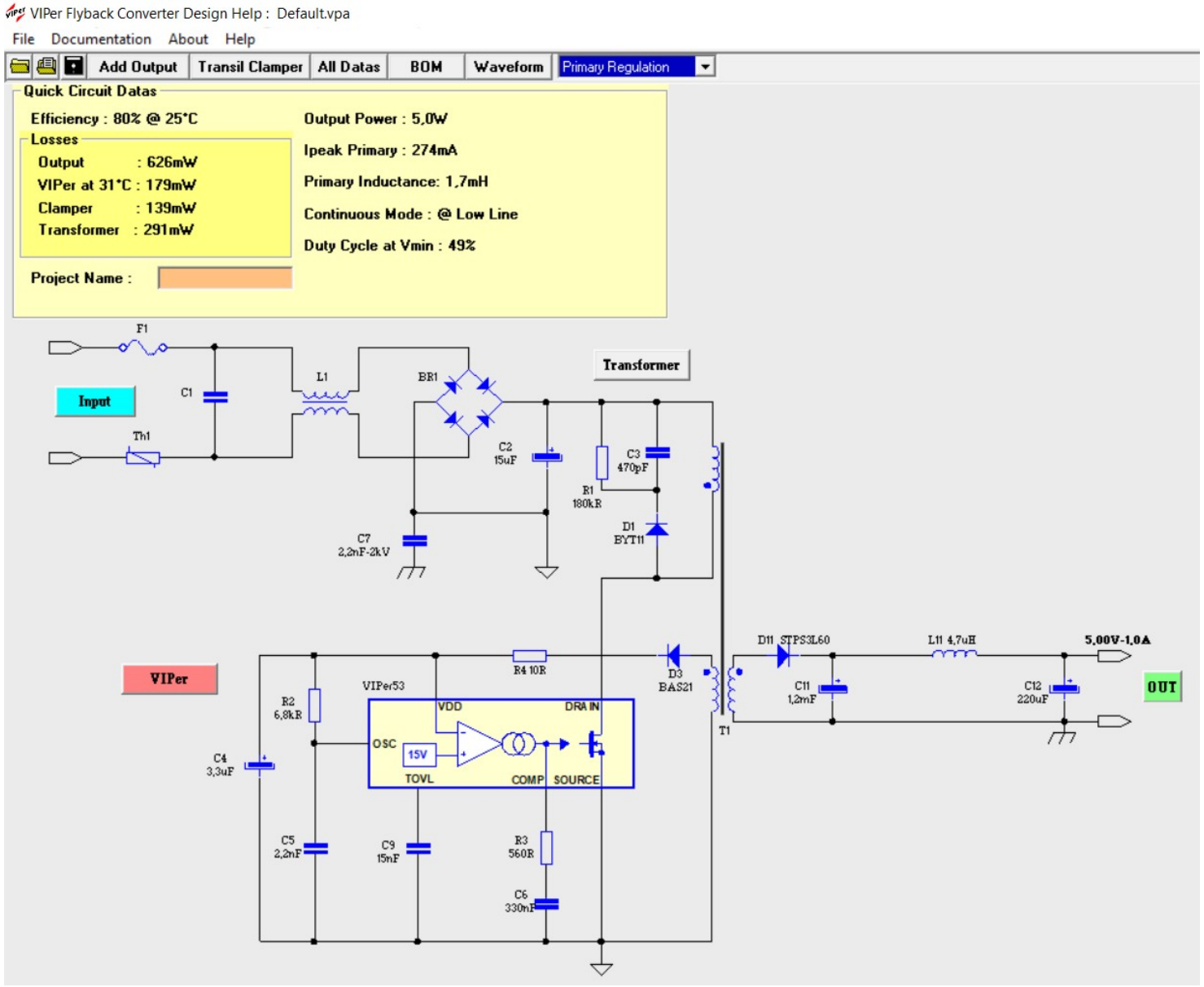
Fonte: Recom Power

Fontes Chaveadas



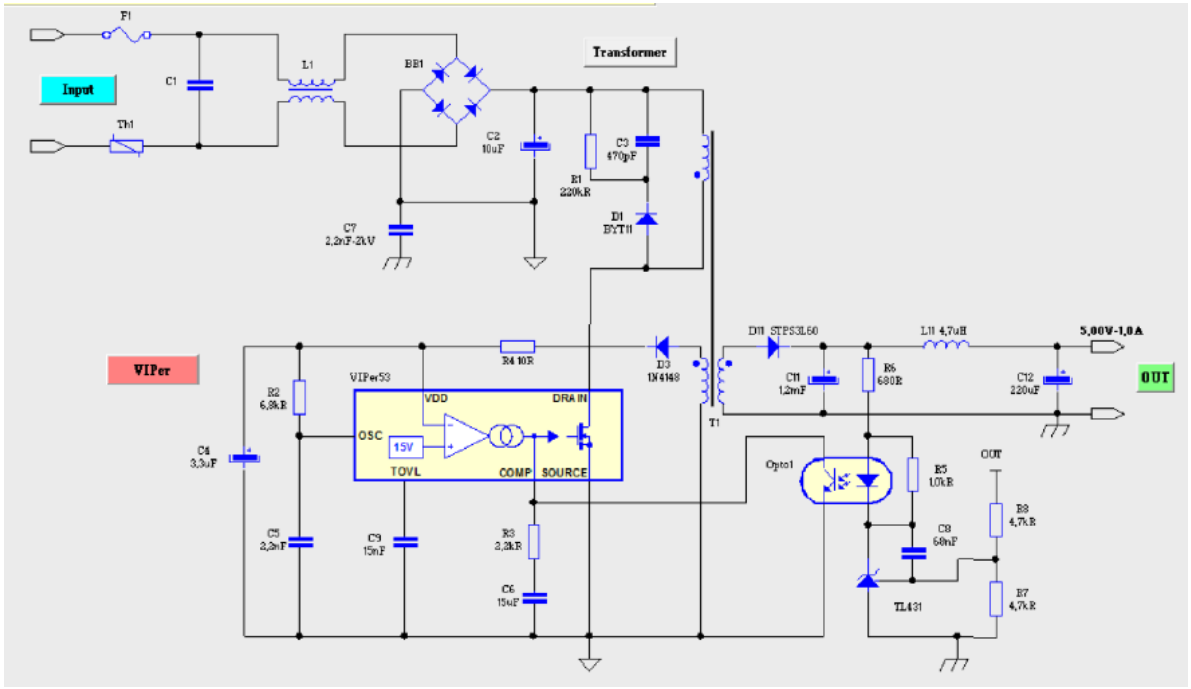
https://my.st.com/analogsimulator/html/#/home

Fontes Chaveadas



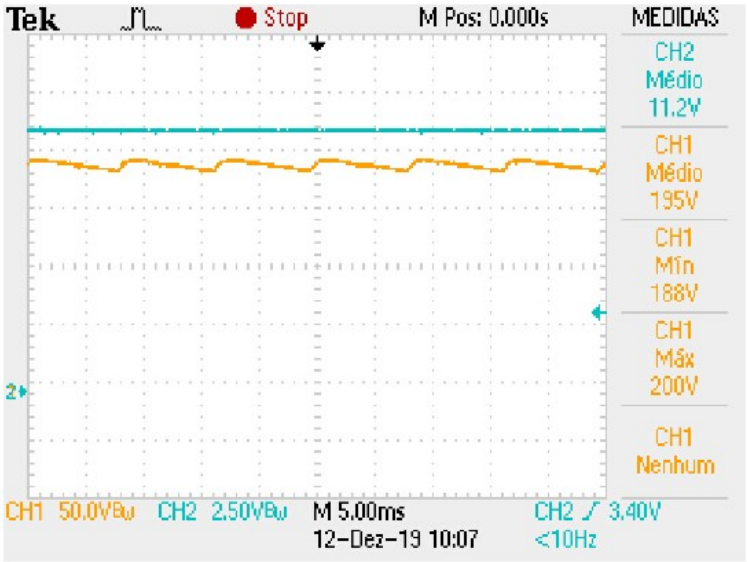
VIPER - ST

Fontes Chaveadas



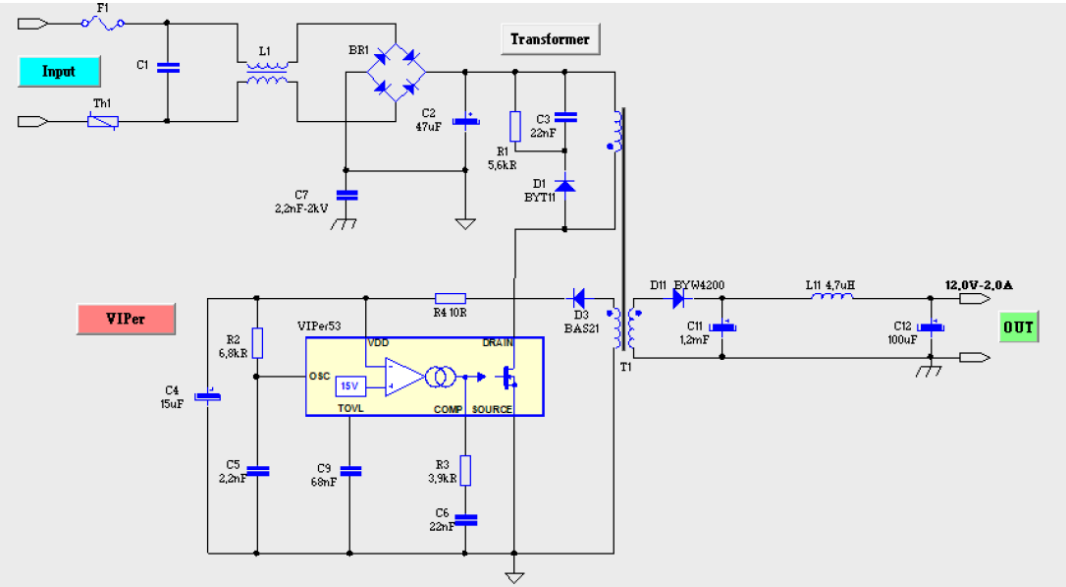
Viper53 – Flyback:
Tensão de Saída: 5V
Corrente Máxima: 1 A



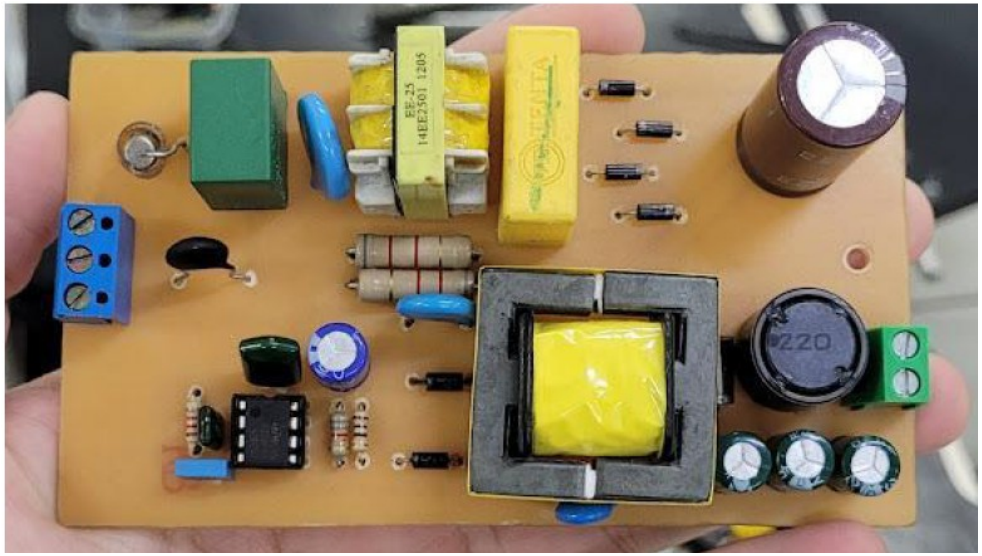


Semestre 2019-2: João Bassani e Ricardo Cirio

Fontes Chaveadas



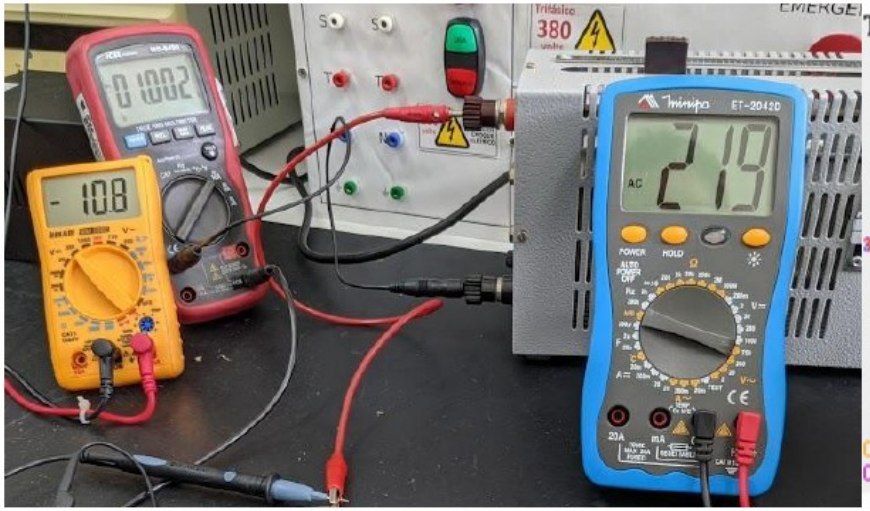
Viper53 – Flyback:
Tensão de Saída:
12V
Corrente Máxima: 2 A

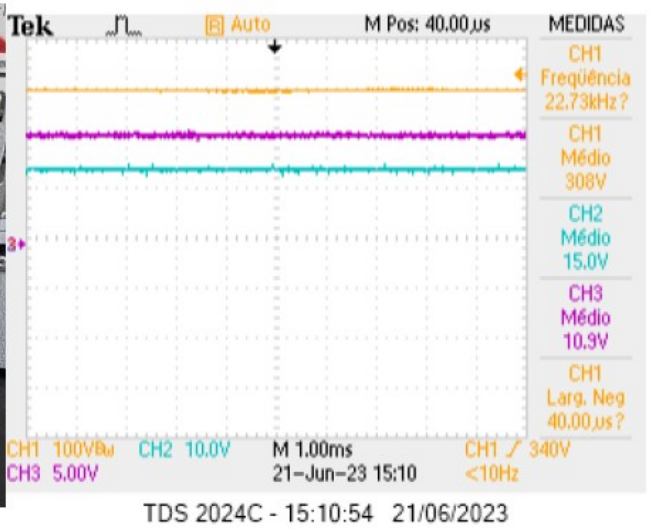


Semestre 2023-1: Gabriel Romero, Jonas Goedert e Matheus Rodrigues

Fontes Chaveadas

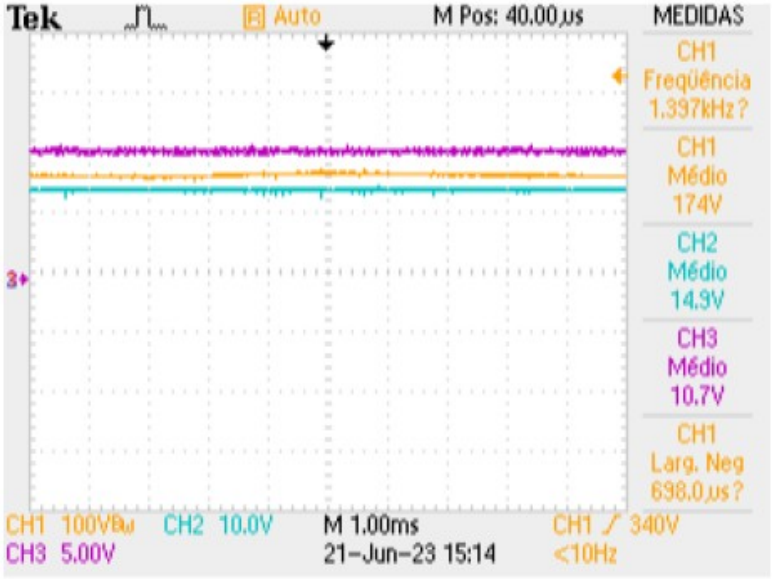
220 V





TDS 2024C - 15:10:54 21/06/2023

127 V

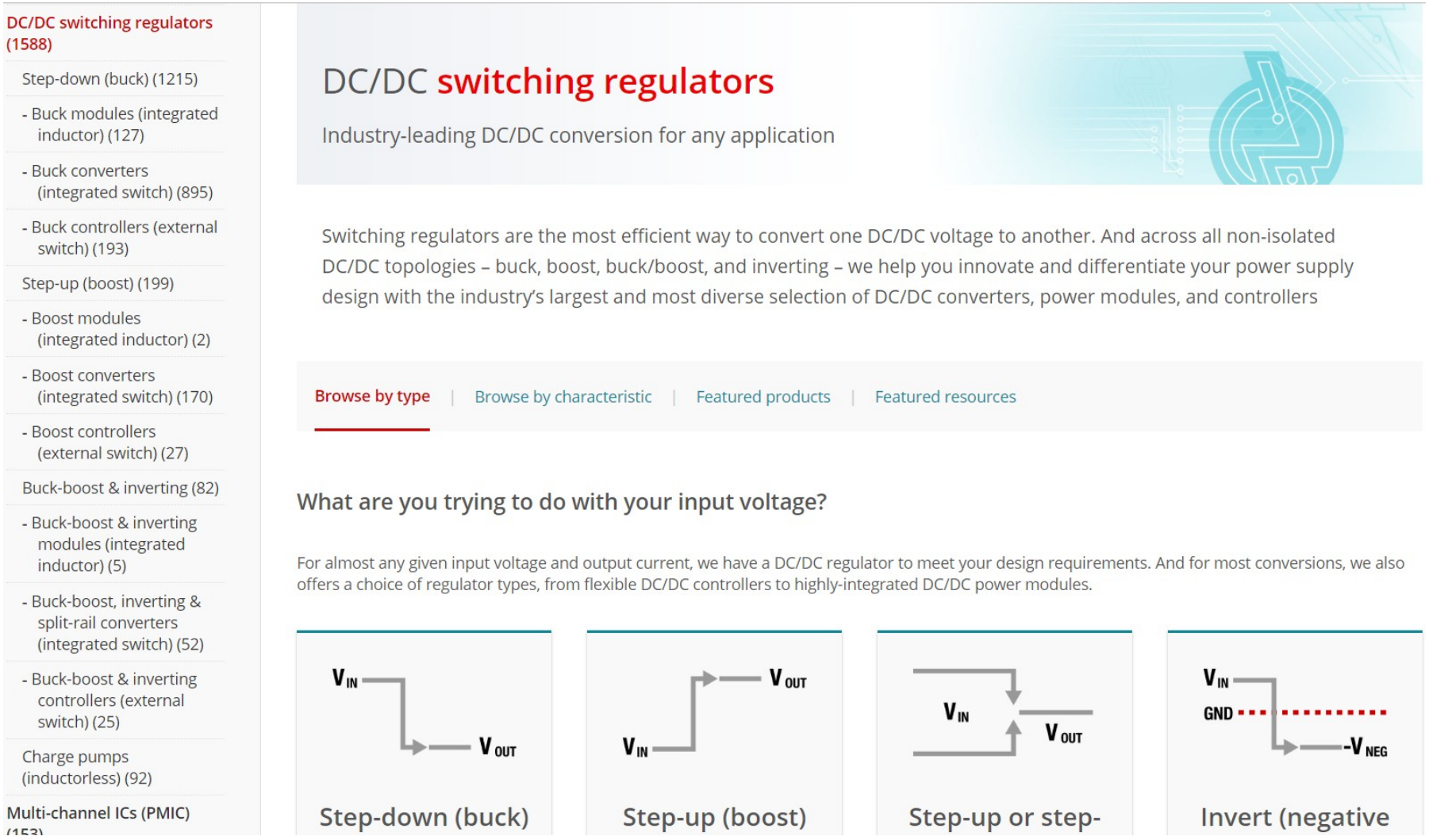


TDS 2024C - 15:15:22 21/06/2023

Semestre 2023-1: Gabriel Romero, Jonas Goedert e Matheus Rodrigues

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http://www.ti.com/power-management/non-isolated-dc-dc-switching-regulators/overview.html

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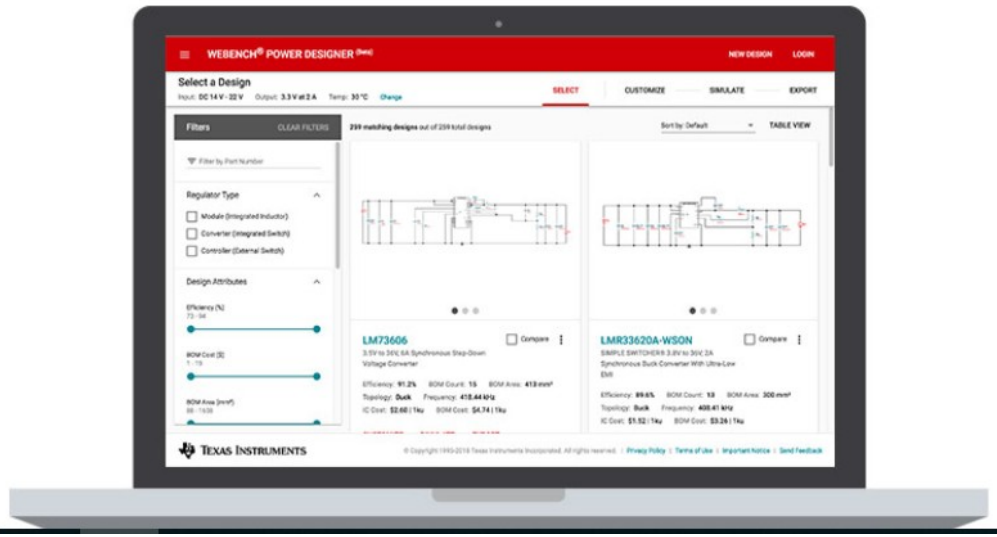
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